The orthoblique rotation for factor analysis

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For the last two decades, the gradient projection algorithm (GPA) has been the most used tool to rotate factor solutions. However, GPA is currently limited to performing either full orthogonal or oblique rotations. Keeping structural zeros in selected positions of the factor correlation matrix after rotation is currently unavailable in Exploratory Factor Analysis. This is a shortcoming because in some cases it is known in advance that some factor correlations must be zero by theory or design. Thereby, we developed a new kind of rotation, the orthoblique rotation, in which both oblique and orthogonal relations between factors coexists. Building on the orthogonal and oblique algorithms of Jennrich, we provide a deeper account of how GPA works and use this account to generalize factor rotation to the orthoblique case. We then demonstrate that this rotation helps to estimate exploratory bi-factor models with multiple general factors in which the general factors may be correlated but must remain orthogonal to the group factors, which can be considered a generalization of the biquartimin and bigeomin criteria of Jennrich and Bentler (2011; 2012). We also illustrate its application to exploratory multitrait-multimethod models, where trait and method factors may correlate within their respective sets but trait–method correlations are constrained to zero by design. Finally, we made available the new orthoblique rotation in the R package `latent`, whose `lrotate()` function serves the same purpose as the `GPArotation` package of Bernaards and Jennrich (2005). Taken together, this work contributes to filling the gap between the confirmatory–exploratory continuum and can be seen as a continuation of the efforts of Jennrich to broaden the flexibility and reach of Exploratory Factor Analysis.

Introduction

Exploratory factor analysis (EFA) is a popular framework for estimating large and complex factor structures. It surpasses the overly restrictive measurement models and the unacceptable model fit indices often encountered in real applications of confirmatory factor analysis [CFA; Marsh et al. (2020)]. In CFA, incorrectly fixing cross-loadings to zero is an important caveat whereas EFA permits the estimation of all the loading parameters of the measurement model by means of a rotation step, reducing the risk of model misspecification.

However, the flexibility of EFA comes at the expense of specifying a suitable simplicity or parsimony criterion to be optimized, so that a factor solution is rotated to a loading matrix that best approximates the desired simple structure. In this regard, the gradient projection algorithm [GPA; Jennrich (2001); Jennrich (2002)] has been the predominant computational tool for rotating loading patterns during the last twenty years. The success of GPA is instantiated by its availability in widely used and reputed statistical programs such as Mplus (Muthén & Muthén, 2017), `lavaan` (Rosseel, 2012), and `psych` (Revelle, 2020). In contrast with previously used rotation procedures like the criterion-specific pairwise algorithms discussed by Jennrich (2001), GPA is simple and elegant. In fact, a general-purpose optimization routine based on GPA can be written in about 50 lines of code in the R language [1]. Moreover, GPA is based on an optimization framework where traditional step-size selection techniques like backtracking line-search and search directions based on first- and second-order derivatives can be used (Liu, 2020).

The generality of GPA for factor rotation was the main contribution of Jennrich (2001) and Jennrich (2002). Rather than developing a different algorithm for every simplicity criterion, they showed that a common computational procedure could optimize many criteria under the same rotation constraints. Bernaards and Jennrich (2005) developed this idea into general-purpose software and separated the evaluation of the criterion and the gradient of the parameters from the operations that enforce the rotation constraints. They also provided derivations, proofs, and software implementation details. Thus, their work provided both the motivation and the computational starting point for the present generalization to a new rotation method. Our purpose is to give a deeper and more explicit geometric account of all the mathematical objects involving factor rotation and the connections between them. Explaining why they are needed, how they are derived from the constraints of the rotation

[1] See the `GPForth` and `GPFoblq` functions from the `GPArotation` R package (Bernaards & Jennrich, 2005) for orthogonal and oblique rotation, respectively

matrix, and how they work together provides a constructive route from the orthogonal and oblique cases to other rotation geometries.

This generalization is very important because in EFA, the applications of GPA are currently limited to either orthogonal and oblique rotation. In other words, all the factors in EFA are currently either constrained to be orthogonal or allowed to correlate. This circumstance is problematic in cases where the researchers know in advance that two sets of factors may be correlated within each set but orthogonal between sets. For example, consider multitrait-multimethod models, in which trait and method factors may correlate within their respective sets but trait–method correlations must be constrained to zero by design. In this scenario, there should be structural zeros in the factor correlation matrix to effectively separate the variance explained by traits and methods. In other cases, it may be desirable to constrain one factor, or a group of factors, to be orthogonal to the remaining factors in order to obtain a more parsimonious pattern structure. For this, consider applications of exploratory bifactor modeling with one or multiple general factors in which all the general factors must be orthogonal to the group factors, but may be correlated among themselves (Jennrich & Bentler, 2012; Jiménez et al., 2023). If the general factors were allowed to correlate with the group factors in an oblique rotation analysis, then the variance explained by each kind of factor could not be correctly disentangled, with the consequence that cross-loadings across the general factors would be biased.

Thereby, a generalization of the oblique and orthogonal rotations that allows for a mix of oblique and orthogonal factors could aid the correct estimation of such factor structures.

The objectives of this work are fourfold. First, we give a coherent and self-contained explanation of EFA and GPA for orthogonal and oblique rotation, connecting the matrix formulas of Jennrich (2001) and Jennrich (2002) with their underlying geometry. Second, using these explanations, we derive the orthoblique rotation, a generalization in which both oblique and orthogonal relations between factors coexists, and show how the conventional rotations are obtained as special cases. Standard errors for the new rotated parameters and correlation-preserving factor scores are also provided. Third, we make this new rotation available in the `latent` R package and explain how the function `lrotate()` extends the rotation framework used in `GPArotation`. Lastly, we illustrate the usefulness of this generalization in two empirical examples concerning exploratory multitrait-multimethod and bifactor models with multiple general factors. In the end, we discuss the benefits of the orthoblique rotation for preserving the flexibility of EFA to uncover cross-loadings and wrong theoretical expectations while

retaining specific factor-correlation restrictions.

Factor rotation

Because rotation is applied only to measurement parameters, without loss of generality we will consider the simplest case of EFA: Exploratory Factor Analysis (EFA). EFA is usually performed in two steps. First, an identified model $\hat{\boldsymbol{\Sigma}} = \boldsymbol{\Lambda}_0\boldsymbol{\Lambda}_0^\top + \boldsymbol{\Theta}$ is estimated by maximum likelihood or least squares by constraining $\boldsymbol{\Lambda}_0$ to have either orthogonal columns or a lower-triangular structure. Second, $\boldsymbol{\Lambda}_0$ is rotated to find an equally fitting model $\hat{\boldsymbol{\Sigma}} = \boldsymbol{\Lambda}\boldsymbol{\Psi}\boldsymbol{\Lambda}^\top + \boldsymbol{\Theta}$, where $\boldsymbol{\Lambda} = \boldsymbol{\Lambda}_0\mathbf{X}^{-\top}$ and $\boldsymbol{\Psi} = \mathbf{X}^\top\mathbf{X}$ is the latent factor correlation matrix. This last step involves finding the optimal rotation matrix $\mathbf{X}$ that minimizes a complexity criterion seeking to obtain a sparse and interpretable $\boldsymbol{\Lambda}$. However, $\mathbf{X}$ must be subject to certain constraints that distinguish orthogonal from oblique rotations. In orthogonal rotation, the constraint is $\mathbf{X}^\top\mathbf{X} = \mathbf{I}$, whereas in oblique rotation it is $\text{diag}(\mathbf{X}^\top\mathbf{X}) = \mathbf{1}$. Therefore, both orthogonal and oblique rotation matrices belong to smooth nonlinear sets of $\mathbb{R}^{n\times n}$ matrices characterized by these specific constraints. Informally, these specific sets are termed manifolds and denoted by $\mathcal{M}$.

The problem of factor rotation is that the optimal orthogonal or oblique rotation matrix $\mathbf{X}$ cannot generally be found by using ordinary optimization algorithms that do not enforce these constraints. For this purpose, the Gradient Projection Algorithm (GPA) has proved to be an elegant and robust solution. GPA builds on differential geometry to perform optimization on nonlinear manifolds. Let $\tilde{f}$ be a complexity function with domain in $\mathbb{R}^{n\times n}$ that we aim to minimize with respect to a rotation matrix $\mathbf{X}$. This function can be any complexity criterion, such as varimax, oblimin, or geomin. Suppose that we want to constrain the solution to lie on a nonlinear manifold $\mathcal{M}$ characterized by a defining smooth function $h : \mathbb{R}^{n\times n} \to \mathbb{R}^d$ that satisfies

$$\mathcal{M} = h^{-1}(\mathbf{0}) = \left\{\mathbf{X} \in \mathbb{R}^{n\times n} : h(\mathbf{X}) = \mathbf{0}\right\},$$

so that any matrix $\mathbf{X} \in \mathcal{M}$ implies $h(\mathbf{X}) = \mathbf{0}$. Here, d is the number of independent constraints defining the manifold. This means that, in the orthogonal case, h collects the $n(n+1)/2$ distinct constraints of the symmetric matrix $\mathbf{X}^\top\mathbf{X}-\mathbf{I}$, whereas in the oblique case it collects the n diagonal constraints. In this manner, the orthogonal and oblique manifolds can

be stated as the sets of full-rank $\mathbf{X}$ matrices satisfying the following conditions, respectively:

$$\mathcal{O}(n) = \left\{\mathbf{X} \in \mathbb{R}^{n\times n} : h(\mathbf{X}) = \mathbf{X}^\top\mathbf{X} - \mathbf{I} = \mathbf{0}\right\},$$
$$\mathcal{OB}(n) = \left\{\mathbf{X} \in \mathbb{R}^{n\times n} : h(\mathbf{X}) = \mathrm{diag}(\mathbf{X}^\top\mathbf{X}) - \mathbf{1} = \mathbf{0}\right\}.$$

To better understand these manifold definitions, an illustration of the orthogonal and oblique manifolds for $n = 3$ is presented in Figure 1. In both manifolds, each column $\mathbf{x}_j$ of the rotation matrix is a unit vector and can therefore be represented on the unit sphere. The oblique manifold is the product of n unit spheres, whereas the orthogonal manifold is the subset of this product in which the vectors are also constrained to point in mutually perpendicular directions.

Orthogonal rotation $\mathcal{O}(3)$ **Oblique rotation** $\mathcal{OB}(3)$

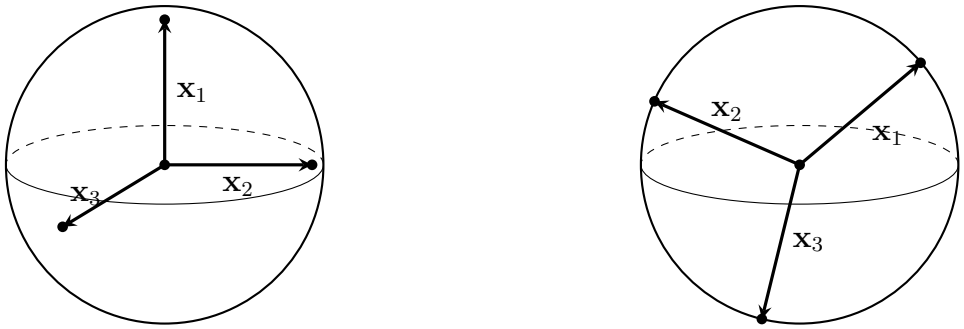


Figure 1: Schematic representation of the orthogonal and oblique manifolds in $n = 3$. In both cases, the columns of $\mathbf{X}$ are unit vectors on a sphere. In $\mathcal{O}(3)$ they are constrained to be orthogonal, whereas in $\mathcal{OB}(3)$ they may point in nonperpendicular directions.

Visually, the GPA consists of moving these vectors across the surface of the unit sphere until the complexity criterion $\tilde{f}$ is minimized, subject to the additional perpendicularity constraints in orthogonal rotation. To move the vectors in this way while minimizing $\tilde{f}$, small steps must be taken through an iterative algorithm. However, this is not trivial. At iteration k, traditional unconstrained optimization algorithms usually compute a new candidate as

$$\mathbf{X}_{k+1} = \mathbf{X}_k + t_k\mathbf{V}_k, \tag{1}$$

where $\mathbf{V}_k$ is a direction along which the complexity function decreases and t_k is a step size. In gradient descent, for example, $\mathbf{V}_k = -\mathrm{grad}\tilde{f}(\mathbf{X}_k)$, where $\mathrm{grad}\tilde{f}(\mathbf{X}_k)$ is the Euclidean

gradient in the unrestricted space $\mathbb{R}^{n\times n}$.[2] The problem is that this updating scheme does not guarantee that the new candidate $\mathbf{X}_{k+1}$ remains on the manifold, not even when $\mathbf{X}_k$ already belongs to it. Furthermore, the descent direction $\mathbf{V}_k$ is also unrestricted in $\mathbb{R}^{n\times n}$ and is not necessarily a valid descent direction on a nonlinear manifold. In other words, a straight-line displacement in the Euclidean space, as shown in the updating Equation 1, is not generally a feasible displacement on a nonlinear manifold.

Hence, the GPA solves two problems: (1) proposing an appropriate descent direction $\mathbf{V}_k$ on $\mathcal{M}$ and (2) securing that, after moving along this direction, the new candidate $\mathbf{X}_{k+1}$ remains on $\mathcal{M}$.

Solution to problem 1: the Riemannian gradient

To address the first problem, we replace the unrestricted Euclidean gradient with its counterpart on the manifold. For the restricted criterion $f = \tilde{f}|_{\mathcal{M}}$, the Riemannian gradient lies in the tangent space and represents how the criterion changes along first-order feasible directions. Its negative is a descent direction whenever the gradient is nonzero. More precisely, it is characterized as follows (Boumal, 2023, Definition 3.58).

Definition 1 *The Riemannian gradient of f at $\mathbf{X} \in \mathcal{M}$, denoted by $gradf(\mathbf{X})$, is the unique element of $\mathcal{T}_{\mathbf{X}}\mathcal{M}$ that satisfies $\mathrm{D}f(\mathbf{X})[\mathbf{V}] = \langle \mathbf{V}, gradf(\mathbf{X})\rangle_{\mathbf{X}}$ for every $\mathbf{V} \in \mathcal{T}_{\mathbf{X}}\mathcal{M}$.*

Informally, if we zoom in enough around a point $\mathbf{X} \in \mathcal{M}$, eventually everything looks linear.[3] Thus, an idea to move over $\mathcal{M}$ consists of finding tangent directions to $\mathbf{X} \in \mathcal{M}$ that look linear from an infinitesimal point of view.

Consider doing a first-order Taylor expansion of h around the current candidate $\mathbf{X}_k$,

$$h(\mathbf{X}_k + t\mathbf{V}_k) \approx h(\mathbf{X}_k) + t\,\mathrm{D}h(\mathbf{X}_k)[\mathbf{V}_k]. \tag{2}$$

Because we can always propose a candidate $\mathbf{X}_k$ that already belongs to $\mathcal{M}$ [4], it follows that $h(\mathbf{X}_k) = \mathbf{0}$. Then, we may choose a $\mathbf{V}_k$ such that $\mathrm{D}h(\mathbf{X}_k)[\mathbf{V}_k] = \mathbf{0}$, so that Equation 2

[2] The Euclidean gradient is characterized by $\mathrm{D}\tilde{f}(\mathbf{X}_k)[\mathbf{V}] = \langle \mathbf{V}, \mathrm{grad}\tilde{f}(\mathbf{X}_k)\rangle$ for any $\mathbf{V} \in \mathbb{R}^{n\times n}$, with $\langle ., .\rangle$ denoting the Frobenius inner product.

[3] As an analogy, the Earth is round but locally everything looks flat.

[4] We can always propose an initial starting point $\mathbf{X}_1 \in \mathcal{M}$.

becomes $\mathbf{0}$ and we preserve the manifold constraints to first order. Therefore, appropriate directions to move along the manifolds constitute the kernel of $\mathrm{D}h(\mathbf{X}_k)$:

$$\mathcal{T}_{\mathbf{X}}\mathcal{M} \equiv \ker \mathrm{D}h(\mathbf{X}) = \left\{\mathbf{V} \in \mathbb{R}^{n\times n} : \mathrm{D}h(\mathbf{X})[\mathbf{V}] = \mathbf{0}\right\},$$

where the subscript k is dropped hereafter for clarity and generality. The set $\mathcal{T}_{\mathbf{X}}\mathcal{M}$ is known as the tangent space of $\mathbf{X}$ on $\mathcal{M}$. In other words, $\mathcal{T}_{\mathbf{X}}\mathcal{M}$ is the set of all infinitesimally feasible directions at $\mathbf{X}$ that are linear. This linearity property is important because it allows traditional optimization algorithms that work in $\mathbb{R}^{n\times n}$ to be used in the GPA framework.

Connecting this new notion of linear space to our previous examples, the tangent spaces of the orthogonal and oblique manifolds at any $\mathbf{X} \in \mathcal{M}$ can be stated as the collection of directions for which the differentials of their respective defining functions are $\mathbf{0}$:

$$\begin{aligned}\mathcal{T}_{\mathbf{X}}\mathcal{O}(n) &= \left\{\mathbf{V} \in \mathbb{R}^{n\times n} : \mathbf{X}^\top\mathbf{V} + \mathbf{V}^\top\mathbf{X} = \mathbf{0}\right\},\\ \mathcal{T}_{\mathbf{X}}\mathcal{OB}(n) &= \left\{\mathbf{V} \in \mathbb{R}^{n\times n} : \mathrm{ddiag}(\mathbf{X}^\top\mathbf{V}) = \mathbf{0}\right\},\end{aligned} \tag{3}$$

where ddiag is an operator that sets the off-diagonal entries of a matrix to zero.

Notice that tangent spaces can be viewed as a collection of the $n^2 - d$ free components in the manifolds, so that $\dim(\mathcal{T}_{\mathbf{X}}\mathcal{O}(n)) = n(n-1)/2$ is the number of free elements in an orthogonal matrix and $\dim(\mathcal{T}_{\mathbf{X}}\mathcal{OB}(n)) = n(n-1)$ is the number of free elements in an oblique matrix. Thereby, we can exploit the above differentials to parameterize the tangent spaces in such a way that the parameterizations preserve the tangent space dimension.

In the orthogonal case, for instance, if we set $\mathbf{V} = \mathbf{X}\boldsymbol{\Omega}$, then by substitution in Equation 3, we have

$$\mathcal{T}_{\mathbf{X}}\mathcal{O}(n) = \left\{\mathbf{X}\boldsymbol{\Omega} : \boldsymbol{\Omega} + \boldsymbol{\Omega}^\top = \mathbf{0}\right\}.$$

This is a valid full-rank parameterization of the tangent space of the orthogonal manifold because $\boldsymbol{\Omega}$ is a skew-symmetric matrix and hence it has $n(n-1)/2$ free elements.

In the oblique case, for a nonsingular rotation matrix $\mathbf{X}$, a useful full-rank parameterization of the tangent space is $\mathbf{V} = \mathbf{X}^{-\top}\boldsymbol{\Omega}$, which by substitution in Equation 3, yields

$$\mathcal{T}_{\mathbf{X}}\mathcal{OB}(n) = \left\{\mathbf{X}^{-\top}\boldsymbol{\Omega} : \mathrm{diag}(\boldsymbol{\Omega}) = \mathbf{0}\right\}.$$

Indeed, $\mathbf{\Omega}$ is an arbitrary off-diagonal matrix with $n(n-1)$ free elements and thus it is a valid full-rank basis of the tangent space of the oblique manifold.

Now, notice that any member of $\mathbb{R}^{n\times n}$ can be decomposed in orthogonal complements. At any point $\mathbf{X} \in \mathcal{M}$, the Euclidean space can be decomposed into the orthogonal direct sum of the tangent and normal spaces,

$$\mathbb{R}^{n\times n} = \mathcal{T}_{\mathbf{X}}\mathcal{M} \oplus \mathcal{N}_{\mathbf{X}}\mathcal{M}. \tag{4}$$

If this $\mathbb{R}^{n\times n}$ element is the Euclidean gradient of $\tilde{f}$ at $\mathbf{X}$, this means that it can be decomposed into an element belonging to the tangent space, $\mathcal{T}_{\mathbf{X}}\mathcal{M}$, and another belonging to the normal space, $\mathcal{N}_{\mathbf{X}}\mathcal{M}$. Therefore, there must exist orthogonal complements satisfying

$$\text{grad}\tilde{f}(\mathbf{X}) = \text{grad}\tilde{f}(\mathbf{X})_{\parallel} + \text{grad}\tilde{f}(\mathbf{X})_{\perp}, \tag{5}$$

where $\text{grad}\tilde{f}(\mathbf{X})_{\parallel}$ and $\text{grad}\tilde{f}(\mathbf{X})_{\perp}$ are the projections of $\text{grad}\tilde{f}(\mathbf{X})$ onto the tangent and normal spaces of the manifold $\mathcal{M}$, respectively (Absil et al., 2009, p. 47). Figure 2 depicts the relationship between these quantities.

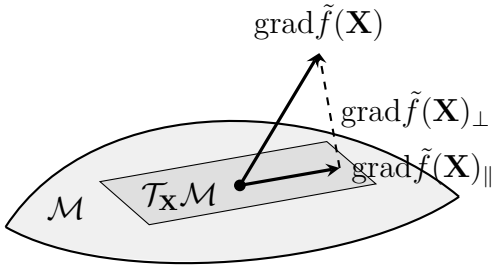


Figure 2: Manifold with the tangent space at $\mathbf{X}$, represented by the black dot. The tangent space is the flat rectangle at $\mathbf{X}$. It can be seen that the Euclidean gradient $\text{grad}\tilde{f}(\mathbf{X})$ can be decomposed in orthogonal complements $\text{grad}\tilde{f}(\mathbf{X})_{\perp}$ and $\text{grad}\tilde{f}(\mathbf{X})_{\parallel}$.

As it can be seen in Figure 2, a tangent space is indeed a linearization of the manifold at a given $\mathbf{X}$ and for any vector in the tangent space there would be another vector from the normal space that is orthogonal to it and satisfies Equation 4. However, to make sense of this notion of orthogonality at any $\mathbf{V} \in \mathcal{T}_{\mathbf{X}}\mathcal{M}$, we must endow the tangent space with a metric like the Frobenius inner product (i.e., $\langle \mathbf{U}, \mathbf{V} \rangle_{\mathbf{X}} = \text{trace}(\mathbf{U}^{\top}\mathbf{V})$), so that $\mathcal{M}$ acquires a Riemannian structure, turning it into a Riemannian manifold. In other words, the collection of inner products on the tangent spaces defines a Riemannian metric on $\mathcal{M}$, turning it into

a Riemannian manifold. And this definition of inner product over the tangent space of $\mathcal{M}$ allows us to develop a definition of orthogonal complements.

Recall that $f = \tilde{f}|_{\mathcal{M}}$ is the restriction of the complexity criterion to the manifold. Using Equation 5 and the Frobenius inner product as Riemannian metric, we arrive at the following identities for any $\mathbf{V} \in \mathcal{T}_{\mathbf{X}}\mathcal{M}$:

$$\begin{aligned}
\left\langle \mathbf{V}, \mathrm{grad}\tilde{f}(\mathbf{X})_{\|} \right\rangle_{\mathbf{X}} &= \mathrm{trace}\left[\mathbf{V}^{\top}\left(\mathrm{grad}\tilde{f}(\mathbf{X}) - \mathrm{grad}\tilde{f}(\mathbf{X})_{\perp}\right)\right] \\
&= \mathrm{trace}\left(\mathbf{V}^{\top}\mathrm{grad}\tilde{f}(\mathbf{X})\right) - \mathrm{trace}\left(\mathbf{V}^{\top}\mathrm{grad}\tilde{f}(\mathbf{X})_{\perp}\right) \\
&= \mathrm{trace}\left(\mathbf{V}^{\top}\mathrm{grad}\tilde{f}(\mathbf{X})\right), \quad \text{because } \mathbf{V} \text{ is orthogonal to } \mathrm{grad}\tilde{f}(\mathbf{X})_{\perp}, \\
&= \mathrm{D}\tilde{f}(\mathbf{X})[\mathbf{V}], \quad \text{by the definition of the Euclidean gradient,} \\
&= \mathrm{D}f(\mathbf{X})[\mathbf{V}], \quad \text{because } f = \tilde{f}|_{\mathcal{M}} \text{ and } \mathbf{V} \in \mathcal{T}_{\mathbf{X}}\mathcal{M} \\
&= \langle \mathbf{V}, \mathrm{grad}f(\mathbf{X}) \rangle_{\mathbf{X}}\,.
\end{aligned}$$

The last equality follows directly from the definition of the Riemannian gradient. And from the preceding identities, this element turns out to be $\mathrm{grad}\tilde{f}(\mathbf{X})_{\|}$, the projection of the Euclidean gradient onto the tangent space of $\mathcal{M}$.

There can be many ways to find the Riemannian gradient, but a good strategy consists of obtaining an expression for $\mathrm{grad}\tilde{f}(\mathbf{X})_{\perp} \in \mathcal{N}_{\mathbf{X}}\mathcal{M}$ and substituting this into Equation 5.

In the case of the orthogonal and oblique manifolds, the parameterizations of the tangent spaces were $\mathbf{X}\boldsymbol{\Omega}$ and $\mathbf{X}^{-\top}\boldsymbol{\Omega}$, respectively, so the normal spaces must be

$$\begin{aligned}
\mathcal{N}_{\mathbf{X}}\mathcal{O}(n) &= \left\{\mathbf{N} : \mathbf{N} = \mathbf{X}\mathbf{A}, \quad \mathbf{A} = \mathbf{A}^{\top}\right\}, \\
\mathcal{N}_{\mathbf{X}}\mathcal{OB}(n) &= \left\{\mathbf{N} : \mathbf{N} = \mathbf{X}\mathbf{A}, \quad \mathbf{A} \text{ is diagonal}\right\},
\end{aligned}$$

respectively. Clearly, the tangent and normal parameterizations are orthogonal in each scenario.[5] Therefore, in both cases the Riemannian gradient can be written as

$$\mathrm{grad}f(\mathbf{X}) = \mathrm{grad}\tilde{f}(\mathbf{X}) - \mathbf{X}\mathbf{A}, \tag{6}$$

where $\mathbf{A}$ must be chosen so that $\mathrm{grad}f(\mathbf{X})$ belongs to the corresponding tangent space.

[5] $\langle \mathbf{X}\boldsymbol{\Omega}, \mathbf{X}\mathbf{A} \rangle = 0$ for a skew-symmetric $\boldsymbol{\Omega}$ and symmetric $\mathbf{A}$ (orthogonal manifold) and $\langle \mathbf{X}^{-\top}\boldsymbol{\Omega}, \mathbf{X}\mathbf{A} \rangle = 0$ for an off-diagonal $\boldsymbol{\Omega}$ and diagonal $\mathbf{A}$ (oblique manifold).

For orthogonal rotation, the Riemannian gradient must satisfy the tangent space equation

$$\mathbf{X}^\top \text{grad} f(\mathbf{X}) + \text{grad} f(\mathbf{X})^\top \mathbf{X} = \mathbf{0}.$$

Substituting $\text{grad} f(\mathbf{X})$ by the orthogonal complements parameterization in Equation 6 gives

$$\begin{aligned}
\mathbf{X}^\top \left(\text{grad}\tilde{f}(\mathbf{X}) - \mathbf{X}\mathbf{A}\right) + \left(\text{grad}\tilde{f}(\mathbf{X}) - \mathbf{X}\mathbf{A}\right)^\top \mathbf{X} &= \mathbf{0} \\
\mathbf{X}^\top \text{grad}\tilde{f}(\mathbf{X}) + \text{grad}\tilde{f}(\mathbf{X})^\top \mathbf{X} - \mathbf{X}^\top\mathbf{X}\mathbf{A} - \mathbf{A}^\top\mathbf{X}^\top\mathbf{X} &= \mathbf{0}, \quad \text{and} \\
\mathbf{X}^\top \text{grad}\tilde{f}(\mathbf{X}) + \text{grad}\tilde{f}(\mathbf{X})^\top \mathbf{X} - 2\mathbf{A} &= \mathbf{0}, \quad \text{because } \mathbf{X}^\top\mathbf{X} = \mathbf{I} \text{ and } \mathbf{A} = \mathbf{A}^\top.
\end{aligned}$$

It then follows that

$$\mathbf{A} = \frac{1}{2}\left[\mathbf{X}^\top \text{grad}\tilde{f}(\mathbf{X}) + \text{grad}\tilde{f}(\mathbf{X})^\top \mathbf{X}\right].$$

Finally, defining $\text{sym}(\mathbf{B}) = \frac{1}{2}\left(\mathbf{B} + \mathbf{B}^\top\right)$, we have $\mathbf{A} = \text{sym}\left(\mathbf{X}^\top \text{grad}\tilde{f}(\mathbf{X})\right)$ and the Riemannian gradient for orthogonal rotation becomes

$$\text{grad} f(\mathbf{X}) = \text{grad}\tilde{f}(\mathbf{X}) - \mathbf{X}\,\text{sym}\left(\mathbf{X}^\top \text{grad}\tilde{f}(\mathbf{X})\right).$$

On the other hand, for oblique rotation, the Riemannian gradient must satisfy the tangent space equation

$$\text{ddiag}\left[\mathbf{X}^\top \text{grad} f(\mathbf{X})\right] = \mathbf{0}.$$

Substituting again $\text{grad} f(\mathbf{X})$ for the orthogonal complements parameterization in Equation 6 gives

$$\begin{aligned}
\text{ddiag}\left[\mathbf{X}^\top \left(\text{grad}\tilde{f}(\mathbf{X}) - \mathbf{X}\mathbf{A}\right)\right] &= \mathbf{0} \\
\text{ddiag}\left[\mathbf{X}^\top \text{grad}\tilde{f}(\mathbf{X})\right] - \text{ddiag}\left(\mathbf{X}^\top\mathbf{X}\mathbf{A}\right) &= \mathbf{0}.
\end{aligned}$$

Because $\mathbf{A}$ is diagonal and $\text{ddiag}\left(\mathbf{X}^\top\mathbf{X}\right) = \mathbf{I}$, we get $\mathbf{A} = \text{ddiag}\left(\mathbf{X}^\top\mathbf{X}\mathbf{A}\right)$.

Therefore,

$$\mathbf{A} = \text{ddiag}\left[\mathbf{X}^\top \text{grad}\tilde{f}(\mathbf{X})\right],$$

and the Riemannian gradient for oblique rotation becomes

$$\text{grad}f(\mathbf{X}) = \text{grad}\tilde{f}(\mathbf{X}) - \mathbf{X}\text{ddiag}\left[\mathbf{X}^\top \text{grad}\tilde{f}(\mathbf{X})\right].$$

These are indeed the same projected gradients formulas denoted by `Gp` in the orthogonal and oblique implementations of Bernaards and Jennrich (pp. 692–695; 2005) and in the `GPArotation` package.

Solution to problem 2: the Retraction

With this generalization of gradients to nonlinear manifolds, we are prepared to compute valid descent directions along the tangent space $\mathcal{T}_{\mathbf{X}}\mathcal{M}$. However, we are not ready to compute valid candidates because the update

$$\mathbf{Z}_{k+1} = \mathbf{X}_k - t_k \text{grad}f(\mathbf{X}_k)$$

does not generally belong to $\mathcal{M}$, however small t_k is. The tangent direction only provides a first-order linear approximation to the manifold at $\mathbf{X}_k$, as we showed before in Equation 2. This means that the condition $\text{D}h(\mathbf{X})[\mathbf{V}] = \mathbf{0}$ does not generally imply that the tangent step $\mathbf{Z} = \mathbf{X} + t\mathbf{V}$ lies exactly on $\mathcal{M}$ for a nonzero t. It only ensures that the constraints are preserved to first order. To see this, let's take the crossproduct of the tangent step $\mathbf{Z}$ to check how it departs from the manifold constraints:

$$\begin{aligned}\mathbf{Z}^\top\mathbf{Z} &= (\mathbf{X} + t\mathbf{V})^\top(\mathbf{X} + t\mathbf{V}) \\ &= \mathbf{X}^\top\mathbf{X} + t(\mathbf{X}^\top\mathbf{V} + \mathbf{V}^\top\mathbf{X}) + t^2\mathbf{V}^\top\mathbf{V}.\end{aligned}$$

In orthogonal rotation, the term $\mathbf{X}^\top\mathbf{V} + \mathbf{V}^\top\mathbf{X}$ vanishes because $\mathbf{V}$ was chosen precisely to satisfy the tangent equation for the orthogonal manifold (Equation 3) and we are left with

$$\mathbf{Z}^\top\mathbf{Z} = \mathbf{I} + t^2\mathbf{V}^\top\mathbf{V}. \tag{7}$$

The orthogonality constraint is therefore violated by the residual $t^2\mathbf{V}^\top\mathbf{V}$. Following the

same argument in the case of the oblique manifold, we have

$$\text{ddiag}(\mathbf{Z}^\top\mathbf{Z}) = \mathbf{I} + t^2\text{ddiag}(\mathbf{V}^\top\mathbf{V}), \tag{8}$$

and the unit norm constraint would be off by a factor of $t^2\text{ddiag}(\mathbf{V}^\top\mathbf{V})$.

Mapping the tangent step back to the manifold is the second problem addressed by the GPA. After taking a step t along the negative Riemannian gradient, the resulting point must be projected back onto the manifold while preserving the initial direction of the tangent step. For this purpose, suppose that we apply some projection $R_{\mathbf{X}} : \mathcal{T}_{\mathbf{X}}\mathcal{M} \to \mathcal{M}$ that restores the constraints exactly while preserving the initial direction. Because the tangent step is already correct to first order, $R_{\mathbf{X}}$ only needs to act on the higher-order residuals. This is exactly what a retraction does: it fixes the constraints of the tangent step without modifying the descent direction (i.e., the negative of the Riemannian gradient in the GPA).

A retraction only needs to satisfy two conditions. The first condition ensures that taking a zero step leaves the current point unchanged. In other words, if we do not move $\mathbf{X}$ along any direction, then the projection must return the same $\mathbf{X}$. The second condition recognizes that the tangent step $\mathbf{X} + t\mathbf{V}$ already satisfies the manifold constraints to first order because $\mathbf{V}$ was picked to satisfy the tangent space equations. Therefore it requires that $R_{\mathbf{X}}(t\mathbf{V})$ has the same initial direction $\mathbf{V}$ as the tangent step. In other words, a retraction restores the manifold constraints exactly without changing the first-order motion on the manifold so that the Riemannian gradient is still the descent direction.

Using these conditions, we can define a retraction in the following manner (Boumal, 2023, Definition 3.47):

Definition 2 *A retraction is a smooth map $R_{\mathbf{X}} : \mathcal{T}_{\mathbf{X}}\mathcal{M} \to \mathcal{M}$ that satisfies the following conditions:*
(1) $R_{\mathbf{X}}(\mathbf{0}) = \mathbf{X}$, and
(2) $DR_{\mathbf{X}}(\mathbf{0})[\mathbf{V}] = \mathbf{V}$ for every $\mathbf{V} \in \mathcal{T}_{\mathbf{X}}\mathcal{M}$.

Combining the Riemannian gradient with a retraction, the valid updating equation that GPA uses becomes

$$\mathbf{X}_{k+1} = R_{\mathbf{X}_k}\left(-t_k\text{grad}f(\mathbf{X}_k)\right). \tag{9}$$

Hence, the Riemannian gradient determines a valid descent direction in the tangent space, whereas the retraction guarantees that the new candidate $\mathbf{X}_{k+1}$ belongs to $\mathcal{M}$ after correcting

the higher-order imprecisions of the tangent step. Figure 3 illustrates how a tangent step is retracted onto the manifold.

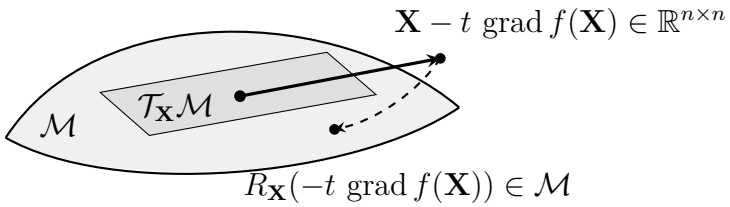


Figure 3: Illustration of a retraction. Starting from $\mathbf{X} \in \mathcal{M}$, a tangent step is taken along the negative Riemannian gradient in the tangent space $\mathcal{T}_{\mathbf{X}}\mathcal{M}$. Since the point $\mathbf{X} - t \text{ grad } f(\mathbf{X})$ does not generally belong to the manifold, the retraction maps it back onto $\mathcal{M}$.

For orthogonal rotation, a suitable retraction is obtained by applying the Gram-Schmidt algorithm to the columns of $\mathbf{Z}$ such that $R_{\mathbf{X}}^{\text{orth}}(t\mathbf{V}) = \mathbf{Q}$, where $\mathbf{Q}$ is the orthonormal matrix of the QR decomposition of the tangent step $\mathbf{Z}$, with the triangular factor chosen to have positive diagonal entries. Alternatively, if $\mathbf{Z} = \mathbf{U}\mathbf{D}\mathbf{W}^\top$ is a singular value decomposition, another valid retraction is $R_{\mathbf{X}}^{\text{orth}}(t\mathbf{V}) = \mathbf{U}\mathbf{W}^\top$. This is actually the orthogonal projection used in the algorithms of Jennrich (2001) and Bernaards and Jennrich (2005). The QR and singular value decomposition retractions need not produce the same finite update, but both are smooth operations that satisfy the conditions in Definition 2, leaving the point $\mathbf{X}$ unchanged when no direction is applied and preserving the tangent direction to first order.

Define the retraction for orthogonal rotation as

$$R_{\mathbf{X}}^{\text{orth}}(t\mathbf{V}) = \mathbf{Z}\left(\mathbf{Z}^\top\mathbf{Z}\right)^{-1/2}, \tag{10}$$

which is equivalent to the singular value decomposition method.

The first condition of Definition 2 is met because at $t = 0$, we have $R_{\mathbf{X}}^{\text{orth}}(\mathbf{0}) = \mathbf{X}$ as $\mathbf{X}$ is already orthonormal. To check the second condition, we can rewrite the retraction in terms of Equation 7 as follows:

$$R_{\mathbf{X}}^{\text{orth}}(t\mathbf{V}) = (\mathbf{X} + t\mathbf{V})\left(\mathbf{I} + t^2\mathbf{V}^\top\mathbf{V}\right)^{-1/2}.$$

Then the derivative of this expression with respect to t becomes

$$\frac{d}{dt}R^{\text{orth}}_{\mathbf{X}}(t\mathbf{V}) = \mathbf{V}\left(\mathbf{I} + t^2\mathbf{V}^\top\mathbf{V}\right)^{-1/2} - t(\mathbf{X} + t\mathbf{V})\left(\mathbf{I} + t^2\mathbf{V}^\top\mathbf{V}\right)^{-3/2}\mathbf{V}^\top\mathbf{V},$$

and evaluated at $t = 0$, we obtain $\frac{d}{dt}R^{\text{orth}}_{\mathbf{X}}(t\mathbf{V})\Big|_{t=0} = \mathbf{V}$ so that the descent direction along the tangent space is preserved.

In case of oblique rotation, a suitable retraction involves normalizing the columns of $\mathbf{Z}$:

$$R^{\text{oblq}}_{\mathbf{X}}(t\mathbf{V}) = \mathbf{Z}\left[\text{ddiag}\left(\mathbf{Z}^\top\mathbf{Z}\right)\right]^{-1/2}, \tag{11}$$

which is the projection used by Jennrich (2002, Appendix A.1).

At $t = 0$ we have $R^{\text{oblq}}_{\mathbf{X}}(\mathbf{0}) = \mathbf{X}$ because the columns of $\mathbf{X}$ are already unit norm. To prove the second condition, we rewrite the normalization using Equation 8 such that

$$R^{\text{oblq}}_{\mathbf{X}}(t\mathbf{V}) = (\mathbf{X} + t\mathbf{V})\left(\mathbf{I} + t^2\,\text{ddiag}(\mathbf{V}^\top\mathbf{V})\right)^{-1/2}.$$

Then the derivative with respect to t is

$$\frac{d}{dt}R^{\text{oblq}}_{\mathbf{X}}(t\mathbf{V}) = \mathbf{V}\left(\mathbf{I} + t^2\,\text{ddiag}(\mathbf{V}^\top\mathbf{V})\right)^{-1/2} - t(\mathbf{X}+t\mathbf{V})\left(\mathbf{I} + t^2\,\text{ddiag}(\mathbf{V}^\top\mathbf{V})\right)^{-3/2}\text{ddiag}(\mathbf{V}^\top\mathbf{V}),$$

which evaluated at $t = 0$ becomes $\frac{d}{dt}R^{\text{oblq}}_{\mathbf{X}}(t\mathbf{V})\Big|_{t=0} = \mathbf{V}$.

Using the definitions of Riemannian gradient and retraction along with their proofs for the orthogonal and oblique cases, we are ready to develop a new rotation: the orthoblique rotation.

The orthoblique manifold

The orthoblique manifold encompasses a mixture of oblique and orthogonal relations between factors. Geometrically, in $n = 3$ it consists of solving the same problem of orthogonal and oblique rotations by minimizing a complexity function while moving vectors on a sphere, with the difference that only one vector is required to be orthogonal to the others (Figure 4).

For clarity, we first consider the case of two blocks of oblique factors and another block

Orthoblique rotation

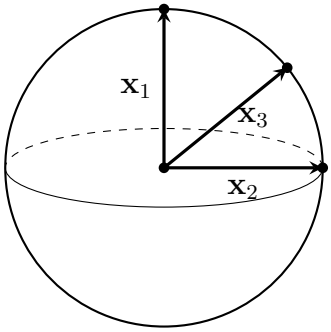


Figure 4: Schematic representation of the orthoblique manifold in $n = 3$. The columns of $\mathbf{X}$ are again unit vectors on a sphere. Here, $\mathbf{x}_1$ is orthogonal to $\mathbf{x}_2$ and $\mathbf{x}_3$, whereas $\mathbf{x}_2$ and $\mathbf{x}_3$ are oblique.

of completely orthogonal factors. However, the following results can be generalized to any number of oblique blocks of factors.

Let $\mathcal{POB}\Big(\{p, q\}, r\Big)$ denote the set of orthoblique matrices in $\mathbb{R}^{n\times n}$ defined by

$$\mathcal{POB}\Big(\{p, q\}, r\Big) = \left\{ \mathbf{X} \in \mathbb{R}^{n\times n} : \mathbf{X}^\top\mathbf{X} = \begin{bmatrix} \mathbf{Y}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{Y}_2 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} \end{bmatrix}, \mathrm{diag}(\mathbf{Y}_1) = \mathbf{1}, \mathrm{diag}(\mathbf{Y}_2) = \mathbf{1} \right\}, \quad (12)$$

where p is the number of oblique factors in the first block, q is the number of oblique factors in the second block, r is the number of completely orthogonal factors in the third block, and thus $n = p + q + r$. Let's partition $\mathbf{X}$ accordingly as $\mathbf{X} = \begin{bmatrix} \mathbf{X}_1 & \mathbf{X}_2 & \mathbf{X}_3 \end{bmatrix}$, where $\mathbf{X}_1 \in \mathbb{R}^{n\times p}$, $\mathbf{X}_2 \in \mathbb{R}^{n\times q}$, and $\mathbf{X}_3 \in \mathbb{R}^{n\times r}$. Then,

$$\mathbf{Y}_1 = \mathbf{X}_1^\top\mathbf{X}_1, \qquad \mathbf{Y}_2 = \mathbf{X}_2^\top\mathbf{X}_2, \qquad \mathbf{I} = \mathbf{X}_3^\top\mathbf{X}_3.$$

Thus, the factors in the oblique blocks $\mathbf{X}_1$ and $\mathbf{X}_2$ are oblique within but orthogonal in between. On the other hand, the factors in $\mathbf{X}_3$ are orthogonal both to each other and to the factors in the two oblique blocks.

The workflow to derive the Riemannian gradient of the orthoblique manifold is as follows: First, we find a smooth defining function h to describe the manifold. Second, we develop a full-rank representation of the tangent and normal spaces in terms of $\mathbf{X}$. Third, using these

parameterizations of the tangent and normal spaces, we find an expression for the projection of the Euclidean gradient of $\tilde{f}$ onto the tangent space. Fourth, we propose a retraction satisfying Definition 2.

Defining function

We can write the orthoblique manifold in terms of a defining function h that collects the constraints in block-matrix form such that $\mathcal{POB}\Big(\{p,q\},r\Big) = \{\mathbf{X} \in \mathbb{R}^{n\times n} : h(\mathbf{X}) = \mathbf{0}\}$, with

$$h(\mathbf{X}) = \begin{bmatrix} \operatorname{ddiag}(\mathbf{X}_1^\top \mathbf{X}_1) - \mathbf{I} & \mathbf{X}_1^\top \mathbf{X}_2 & \mathbf{X}_1^\top \mathbf{X}_3 \\ \mathbf{X}_2^\top \mathbf{X}_1 & \operatorname{ddiag}(\mathbf{X}_2^\top \mathbf{X}_2) - \mathbf{I} & \mathbf{X}_2^\top \mathbf{X}_3 \\ \mathbf{X}_3^\top \mathbf{X}_1 & \mathbf{X}_3^\top \mathbf{X}_2 & \mathbf{X}_3^\top \mathbf{X}_3 - \mathbf{I} \end{bmatrix}.$$

As already explained, h is understood to collect only the independent entries of this symmetric block matrix: the p and q diagonal constraints in the first two blocks, the entries in the three upper off-diagonal blocks, and the $r(r+1)/2$ distinct entries of the third diagonal block. Hence, the number of independent constraints is $d = p + q + pq + pr + qr + r(r+1)/2$.

Tangent space

The tangent space is obtained by differentiating the previous block-matrix representation of $\mathcal{POB}\Big(\{p,q\},r\Big)$. First, let $\mathbf{V}$ be a tangent matrix with partitions $\mathbf{V} = \begin{bmatrix} \mathbf{V}_1 & \mathbf{V}_2 & \mathbf{V}_3 \end{bmatrix}$. Then, the tangent space can be represented as

$$\mathcal{T}_{\mathbf{X}}\mathcal{POB}\Big(\{p,q\},r\Big) = \left\{ \mathbf{V} \in \mathbb{R}^{n\times n} : \begin{bmatrix} \operatorname{ddiag}(\mathbf{X}_1^\top \mathbf{V}_1) & \mathbf{X}_1^\top \mathbf{V}_2 + \mathbf{V}_1^\top \mathbf{X}_2 & \mathbf{X}_1^\top \mathbf{V}_3 + \mathbf{V}_1^\top \mathbf{X}_3 \\ \mathbf{X}_2^\top \mathbf{V}_1 + \mathbf{V}_2^\top \mathbf{X}_1 & \operatorname{ddiag}(\mathbf{X}_2^\top \mathbf{V}_2) & \mathbf{X}_2^\top \mathbf{V}_3 + \mathbf{V}_2^\top \mathbf{X}_3 \\ \mathbf{X}_3^\top \mathbf{V}_1 + \mathbf{V}_3^\top \mathbf{X}_1 & \mathbf{X}_3^\top \mathbf{V}_2 + \mathbf{V}_3^\top \mathbf{X}_2 & \mathbf{X}_3^\top \mathbf{V}_3 + \mathbf{V}_3^\top \mathbf{X}_3 \end{bmatrix} = \mathbf{0} \right\}. \tag{13}$$

At a nonsingular rotation matrix $\mathbf{X}$, we can parameterize this tangent space as $\mathbf{V} = \mathbf{X}^{-\top}\mathbf{\Omega}$, where $\mathbf{\Omega} = \mathbf{X}^\top \mathbf{V}$. Then, by substitution we have

$$\mathcal{T}_{\mathbf{X}}\mathcal{POB}\Big(\{p,q\},r\Big) = \left\{ \mathbf{X}^{-\top}\mathbf{\Omega} : \begin{bmatrix} \operatorname{ddiag}(\mathbf{\Omega}_{11}) & \mathbf{\Omega}_{12} + \mathbf{\Omega}_{21}^\top & \mathbf{\Omega}_{13} + \mathbf{\Omega}_{31}^\top \\ \mathbf{\Omega}_{21} + \mathbf{\Omega}_{12}^\top & \operatorname{ddiag}(\mathbf{\Omega}_{22}) & \mathbf{\Omega}_{23} + \mathbf{\Omega}_{32}^\top \\ \mathbf{\Omega}_{31} + \mathbf{\Omega}_{13}^\top & \mathbf{\Omega}_{32} + \mathbf{\Omega}_{23}^\top & \mathbf{\Omega}_{33} + \mathbf{\Omega}_{33}^\top \end{bmatrix} = \mathbf{0} \right\},$$

where $\mathbf{\Omega}$ has the following block structure:

$$\mathbf{\Omega} = \begin{bmatrix} \mathbf{\Omega}_{11} & \mathbf{\Omega}_{12} & \mathbf{\Omega}_{13} \\ -\mathbf{\Omega}_{12}^\top & \mathbf{\Omega}_{22} & \mathbf{\Omega}_{23} \\ -\mathbf{\Omega}_{13}^\top & -\mathbf{\Omega}_{23}^\top & \mathbf{\Omega}_{33} \end{bmatrix}.$$

$\mathbf{\Omega}_{11}$ and $\mathbf{\Omega}_{22}$ can be arbitrary zero-diagonal matrices and $\mathbf{\Omega}_{33}$ must be skew-symmetric. The block matrices in the upper diagonal can be totally arbitrary, but the must be skew-symmetric with respect to the lower diagonal blocks. We can see that $\dim(\mathbf{\Omega}) = p(p-1) + q(q-1) + pq + pr + qr + r(r-1)/2$. Indeed, this is the same than $n^2 - d$ and hence $\mathbf{X}^{-\top}\mathbf{\Omega}$ is a valid full-rank basis of the tangent space.

Normal space

Following the same principle as for the orthogonal and oblique manifolds, the normal space is the orthogonal complement of the tangent space. A valid parameterization can be constructed by stacking the orthogonal complements of the individual blocks of $\mathbf{\Omega}$. We already showed that any diagonal matrix is an orthogonal complement of any off-diagonal matrix and that any symmetric matrix is an orthogonal complement of any skew-symmetric matrix. Therefore,

$$\mathcal{N}_{\mathbf{X}}\mathcal{POB}\Big(\{p,q\},r\Big) = \left\{ \mathbf{N} = \mathbf{XA} : \mathbf{A} = \begin{bmatrix} \mathbf{D}_1 & \mathbf{B}_{12} & \mathbf{B}_{13} \\ \mathbf{B}_{12}^\top & \mathbf{D}_2 & \mathbf{B}_{23} \\ \mathbf{B}_{13}^\top & \mathbf{B}_{23}^\top & \mathbf{S} \end{bmatrix} \right\},$$

where $\mathbf{D}_1$ and $\mathbf{D}_2$ are diagonal matrices, $\mathbf{S}$ is symmetric, and $\mathbf{B}_{12}$, $\mathbf{B}_{13}$, and $\mathbf{B}_{23}$ are unrestricted matrices with transposed counterparts below the diagonal.

Indeed, if $\mathbf{V} = \mathbf{X}^{-\top}\mathbf{\Omega} \in \mathcal{T}_{\mathbf{X}}\mathcal{POB}\Big(\{p,q\},r\Big)$ and $\mathbf{N} = \mathbf{XA} \in \mathcal{N}_{\mathbf{X}}\mathcal{POB}\Big(\{p,q\},r\Big)$, then

$$\begin{aligned} \langle \mathbf{V}, \mathbf{N} \rangle &= \operatorname{trace}\Big(\mathbf{V}^\top \mathbf{N}\Big) \\ &= \operatorname{trace}\Big(\mathbf{\Omega}^\top \mathbf{X}^{-1}\mathbf{XA}\Big) \\ &= \operatorname{trace}\Big(\mathbf{\Omega}^\top \mathbf{A}\Big) \\ &= 0, \end{aligned}$$

and the next step is to find the right $\mathbf{A}$ to construct an expression of the Riemannian gradient.

Riemannian gradient

For clarity, let $\mathbf{G} = \text{grad}\tilde{f}(\mathbf{X})$ denote the Euclidean gradient. As in the orthogonal and oblique cases, the Riemannian gradient here is obtained by subtracting its normal component (Equation 6), leaving $\text{grad}f(\mathbf{X}) = \mathbf{G} - \mathbf{X}\mathbf{A}$. The matrix $\mathbf{A}$ must be chosen so that $\text{grad}f(\mathbf{X}) \in \mathcal{T}_{\mathbf{X}}\mathcal{POB}$. Hence, substituting $\mathbf{V}$ for this parameterization of the Riemannian gradient in the equation for the tangent space, all the components in the following symmetric matrix become $\mathbf{0}$:

$$\begin{bmatrix} \text{ddiag}\left(\mathbf{X}_1^\top\mathbf{G}_1 - \mathbf{Y}_1\mathbf{D}_1\right) & \mathbf{X}_1^\top\mathbf{G}_2 + \mathbf{G}_1^\top\mathbf{X}_2 - \mathbf{Y}_1\mathbf{B}_{12} - \mathbf{B}_{12}\mathbf{Y}_2 & \mathbf{X}_1^\top\mathbf{G}_3 + \mathbf{G}_1^\top\mathbf{X}_3 - \mathbf{Y}_1\mathbf{B}_{13} - \mathbf{B}_{13} \\ & \text{ddiag}\left(\mathbf{X}_2^\top\mathbf{G}_2 - \mathbf{Y}_2\mathbf{D}_2\right) & \mathbf{X}_2^\top\mathbf{G}_3 + \mathbf{G}_2^\top\mathbf{X}_3 - \mathbf{Y}_2\mathbf{B}_{23} - \mathbf{B}_{23} \\ & & \mathbf{X}_3^\top\mathbf{G}_3 + \mathbf{G}_3^\top\mathbf{X}_3 - 2\mathbf{S} \end{bmatrix}.$$

Each block can now be solved separately.[6] Furthermore, the diagonal blocks of the above expression are the same equations as in the oblique and orthogonal demonstrations but represented in block format. Thereby, the unique $\mathbf{A}$ that satisfies Equation 6 becomes

$$\mathbf{A} = \begin{bmatrix} \mathbf{D}_1 = \text{ddiag}\left(\mathbf{X}_1^\top\mathbf{G}_1\right) & \mathbf{B}_{12} & \mathbf{B}_{13} \\ \mathbf{B}_{12}^\top & \mathbf{D}_2 = \text{ddiag}\left(\mathbf{X}_2^\top\mathbf{G}_2\right) & \mathbf{B}_{23} \\ \mathbf{B}_{13}^\top & \mathbf{B}_{23}^\top & \mathbf{S} = \text{sym}(\mathbf{X}_3^\top\mathbf{G}_3) \end{bmatrix}.$$

For the submatrices at the intersection of the oblique and orthogonal blocks, the solutions are

$$\mathbf{B}_{13} = (\mathbf{Y}_1 + \mathbf{I})^{-1}(\mathbf{X}_1^\top\mathbf{G}_3 + \mathbf{G}_1^\top\mathbf{X}_3),$$
$$\mathbf{B}_{23} = (\mathbf{Y}_2 + \mathbf{I})^{-1}(\mathbf{X}_2^\top\mathbf{G}_3 + \mathbf{G}_2^\top\mathbf{X}_3).$$

Finally, the only remaining block is the intersection between the two oblique blocks, where

$$\mathbf{Y}_1\mathbf{B}_{12} + \mathbf{B}_{12}\mathbf{Y}_2 = \mathbf{X}_1^\top\mathbf{G}_2 + \mathbf{G}_1^\top\mathbf{X}_2.$$

This is a Sylvester equation. Let $\mathbf{Q} = \mathbf{X}_1^\top\mathbf{G}_2 + \mathbf{G}_1^\top\mathbf{X}_2$, $\mathbf{Y}_1 = \mathbf{U}_1\mathbf{V}_1\mathbf{U}_1^\top$, and $\mathbf{Y}_2 = \mathbf{U}_2\mathbf{V}_2\mathbf{U}_2^\top$, where the last two expressions are the eigendecompositions of the symmetric correlation matrices $\mathbf{Y}_1$ and $\mathbf{Y}_2$, respectively. Then, by substitution we have

[6] Notice that we omitted the lower triangle of the matrix because it is redundant with the transpose of the upper triangle.

$$(\mathbf{U}_1\mathbf{V}_1\mathbf{U}_1^\top)\mathbf{B}_{12} + \mathbf{B}_{12}(\mathbf{U}_2\mathbf{V}_2\mathbf{U}_2^\top) = \mathbf{Q}.$$

Using the orthonormality property of the eigenvectors, multiplying the equation on the left by $\mathbf{U}_1^\top$ and on the right by $\mathbf{U}_2$, we get

$$\mathbf{V}_1(\mathbf{U}_1^\top\mathbf{B}_{12}\mathbf{U}_2) + (\mathbf{U}_1^\top\mathbf{B}_{12}\mathbf{U}_2)\mathbf{V}_2 = \mathbf{U}_1^\top\mathbf{Q}\mathbf{U}_2.$$

Let $\mathbf{W} = \mathbf{U}_1^\top\mathbf{B}_{12}\mathbf{U}_2$ so that the expression above can be written as $\mathbf{V}_1\mathbf{W} + \mathbf{W}\mathbf{V}_2 = \mathbf{U}_1^\top\mathbf{Q}\mathbf{U}_2$. Because $\mathbf{V}_1$ and $\mathbf{V}_2$ are diagonal, this equation can be solved elementwise as

$$w_{ij} = \frac{(\mathbf{U}_1^\top\mathbf{Q}\mathbf{U}_2)_{ij}}{(\mathbf{V}_1)_{ii} + (\mathbf{V}_2)_{jj}}.$$

Finally, multiplying again the left and right sides by $\mathbf{U}_1$ and $\mathbf{U}_2^\top$, respectively, the solution becomes

$$\mathbf{B}_{12} = \mathbf{U}_1\mathbf{W}\mathbf{U}_2^\top.$$

Retraction

After computing the Riemannian gradient, the remaining problem is to map $\mathbf{Z} = \mathbf{X} - t\mathrm{grad}f(\mathbf{X})$ back onto the orthoblique manifold. For this, we need to find a function $R_{\mathbf{X}}^{\mathcal{POB}} : \mathcal{T}_{\mathbf{X}}\mathcal{POB} \to \mathcal{POB}$ that satisfies the conditions of Definition 2.

Let $\mathbf{R}$ be the retracted tangent step $\mathbf{R} = R_{\mathbf{X}}^{\mathcal{POB}}(t\mathbf{V})$. Let us partition the matrices $\mathbf{Z}$ and $\mathbf{R}$ into blocks $\mathbf{Z} = \begin{bmatrix}\mathbf{Z}_1 & \mathbf{Z}_2 & \mathbf{Z}_3\end{bmatrix}$ and $\mathbf{R} = \begin{bmatrix}\mathbf{R}_1 & \mathbf{R}_2 & \mathbf{R}_3\end{bmatrix}$ such that the submatrices indexed by 1 and 2 correspond to the first two oblique blocks, respectively, and those indexed by 3 correspond to the block of orthogonal factors.

The retraction of the first oblique block can be simply the retraction of $\mathbf{Z}_1$ onto the space set of normalized columns,

$$\mathbf{R}_1 = \mathbf{Z}_1\left[\mathrm{ddiag}\left(\mathbf{Z}_1^\top\mathbf{Z}_1\right)\right]^{-1/2},$$

so that $\mathrm{diag}(\mathbf{R}_1^\top\mathbf{R}_1) = \mathbf{1}$.

The second oblique block also has to be made orthogonal to the first block. To do this, we first project $\mathbf{Z}_2$ onto the orthogonal space of $\mathbf{Z}_1$ before normalizing the result. First, define the orthogonal projector onto the column space of $\mathbf{R}_1$ as $\mathbf{P}_1 = \mathbf{R}_1\left(\mathbf{R}_1^\top\mathbf{R}_1\right)^{-1}\mathbf{R}_1^\top$. Then,

$\widetilde{\mathbf{Z}}_2 = (\mathbf{I} - \mathbf{P}_1)\,\mathbf{Z}_2$ is orthogonal to $\mathbf{R}_1$. Second, the columns are subsequently normalized to obtain

$$\mathbf{R}_2 = \widetilde{\mathbf{Z}}_2 \left[\operatorname{ddiag}\left(\widetilde{\mathbf{Z}}_2^\top \widetilde{\mathbf{Z}}_2\right)\right]^{-1/2}.$$

Consequently, $\mathbf{R}_1^\top \mathbf{R}_2 = \mathbf{0}$ and $\operatorname{diag}\left(\mathbf{R}_2^\top \mathbf{R}_2\right) = \mathbf{1}$.

Finally, to get $\mathbf{R}_3$, combine the two oblique blocks columnwise in $\mathbf{R}_{12} = \begin{bmatrix}\mathbf{R}_1 & \mathbf{R}_2\end{bmatrix}$ and define the projector onto their joint column space, $\mathbf{P}_{12} = \mathbf{R}_{12}\left(\mathbf{R}_{12}^\top \mathbf{R}_{12}\right)^{-1}\mathbf{R}_{12}^\top$. As before, the orthogonal block is first projected onto the orthogonal complement of both oblique blocks to yield $\widetilde{\mathbf{Z}}_3 = (\mathbf{I} - \mathbf{P}_{12})\,\mathbf{Z}_3$ and its columns are then orthonormalized using

$$\mathbf{R}_3 = \widetilde{\mathbf{Z}}_3 \left(\widetilde{\mathbf{Z}}_3^\top \widetilde{\mathbf{Z}}_3\right)^{-1/2}$$

As a result, $\mathbf{R}_1^\top \mathbf{R}_3 = \mathbf{0}$, $\mathbf{R}_2^\top \mathbf{R}_3 = \mathbf{0}$, $\mathbf{R}_3^\top \mathbf{R}_3 = \mathbf{I}$ and $\mathbf{R}_3$ becomes completely orthogonal.

Taken together, the following blockwise construction defines the retracted orthoblique update

$$R_{\mathbf{X}}^{\mathcal{POB}}(t\mathbf{V}) = \begin{bmatrix}\mathbf{R}_1 & \mathbf{R}_2 & \mathbf{R}_3\end{bmatrix}.$$

Notice that we already proved that the orthonormalization (10) and normalization (11) operations are valid retractions for the orthogonal and oblique manifolds, respectively. To show that $R_{\mathbf{X}}^{\mathcal{POB}}$ is another valid retraction for the orthoblique manifold, we just need to show that the projection of rotation blocks onto the orthogonal space of previous rotation blocks also satisfies the retraction conditions.

First, define $P_{\mathbf{X}_j}$ to be a function that takes a block j and projects it onto the orthogonal space of a previous retracted block i:

$$\begin{aligned} P_{\mathbf{X}_j}(t\mathbf{V}_j) &= (\mathbf{I} - \mathbf{R}_i(\mathbf{R}_i^\top \mathbf{R}_i)^{-1}\mathbf{R}_i^\top)\mathbf{Z}_j \\ &= \mathbf{Z}_j - \mathbf{R}_i(\mathbf{R}_i^\top \mathbf{R}_i)^{-1}\mathbf{R}_i^\top \mathbf{Z}_j. \end{aligned} \tag{14}$$

At $t = 0$, $\mathbf{R}_i$ reduces to $\mathbf{X}_i$ and $\mathbf{Z}_j$ reduces to $\mathbf{X}_j$. Since $\mathbf{X}_i^\top \mathbf{X}_j = \mathbf{0}$, the second term of this expression vanishes and we are left with $P_{\mathbf{X}_j}(t\mathbf{V}_j)\Big|_{t=0} = \mathbf{X}_j$. Therefore, $R_{\mathbf{X}}^{\mathcal{POB}}(\mathbf{0}) = \mathbf{X}$, satisfying the first retraction condition.

To proof the second retraction condition we first differentiate Equation 14 using the product

rule,

$$\begin{aligned} DP_{\mathbf{X}_j}(t\mathbf{V}_j) = \mathbf{V}_j &- (D\mathbf{R}_i)(\mathbf{R}_i^\top \mathbf{R}_i)^{-1}\mathbf{R}_i^\top \mathbf{Z}_j \\ &- \mathbf{R}_i D\left[(\mathbf{R}_i^\top \mathbf{R}_i)^{-1}\right]\mathbf{R}_i^\top \mathbf{Z}_j \\ &- \mathbf{R}_i(\mathbf{R}_i^\top \mathbf{R}_i)^{-1}(D\mathbf{R}_i)^\top \mathbf{Z}_j \\ &- \mathbf{R}_i(\mathbf{R}_i^\top \mathbf{R}_i)^{-1}\mathbf{R}_i^\top \mathbf{V}_j. \end{aligned}$$

Next, we evaluate this expression at $t = 0$. Again, since $\mathbf{R}_i$ reduces to $\mathbf{X}_i$ and $\mathbf{Z}_j$ reduces to $\mathbf{X}_j$, $\mathbf{X}_i^\top \mathbf{X}_j = \mathbf{0}$ and the two terms containing $\mathbf{R}_i^\top \mathbf{Z}_j$ vanish. Therefore,

$$DP_{\mathbf{X}_j}(t\mathbf{V}_j)\Big|_{t=0} = \mathbf{V}_j - \mathbf{X}_i(\mathbf{X}_i^\top \mathbf{X}_i)^{-1}\left[D\mathbf{R}_i^\top\Big|_{t=0}\mathbf{X}_j + \mathbf{X}_i^\top \mathbf{V}_j\right].$$

If the previous projected block $\mathbf{R}_i$ already satisfies the second retraction condition, then $D\mathbf{R}_i|_{t=0} = \mathbf{V}_i$ and we obtain

$$DP_{\mathbf{X}_j}(t\mathbf{V}_j)\Big|_{t=0} = \mathbf{V}_j - \mathbf{X}_i(\mathbf{X}_i^\top \mathbf{X}_i)^{-1}\left(\mathbf{V}_i^\top \mathbf{X}_j + \mathbf{X}_i^\top \mathbf{V}_j\right).$$

But the tangent-space equations (13) states that $\mathbf{V}_i^\top \mathbf{X}_j + \mathbf{X}_i^\top \mathbf{V}_j = \mathbf{0}$. Hence, $DP_{\mathbf{X}_j}(t\mathbf{V}_j)\Big|_{t=0} = \mathbf{V}_j$.

Finally, because we already showed that doing normalization or orthogonalization preserve first-order directions, it follows that applying these operations afterwards still preserve this direction and hence $DR_{\mathbf{X}}^{\mathcal{POB}}(t\mathbf{V})\Big|_{t=0} = \mathbf{V}$.

Notice that any block order also satisfies these retraction conditions, provided each block is projected onto the orthogonal complement of all previously processed blocks and then normalized if they are oblique or orthonormalized if it is orthogonal. Different orders may produce different finite updates but they always preserve the same first-order direction.

Generalization to an arbitrary number of oblique blocks

The previous construction of the orthoblique manifold can be extended to any number m of oblique blocks. Let p_i be the number of factors in the ith oblique block and let r be the number of completely orthogonal factors, so that $n = \sum_{i=1}^{m} p_i + r$. Then,

$$\mathcal{POB}\Big(\{p_1,\ldots,p_m\},r\Big)=\left\{\mathbf{X}\in\mathbb{R}^{n\times n}:\mathbf{X}^\top\mathbf{X}=\begin{bmatrix}\mathbf{Y}_1 & \mathbf{0} & \ldots & \mathbf{0}\\ \vdots & \ddots & \mathbf{0} & \vdots\\ \vdots & \mathbf{0} & \mathbf{Y}_m & \mathbf{0}\\ \mathbf{0} & \ldots & \mathbf{0} & \mathbf{I}\end{bmatrix},\quad \operatorname{diag}(\mathbf{Y}_i)=\mathbf{1}\ \forall i\right\}.$$

In this setup, the tangent space, normal space, and retraction would involve additional oblique blocks, pairs of oblique blocks, and pairs of oblique and orthogonal blocks. But all the formulas of these quantities follow by applying the same blockwise approach as before.

Furthermore, this generalization shows that orthogonal and oblique rotations are special cases of the orthoblique manifold. If there are no oblique blocks and $r = n$, then $\mathcal{POB}(\varnothing, n) = \mathcal{O}(n)$. And if there is one oblique block with $p_1 = n$ and $r = 0$, then $\mathcal{POB}(\{n\}, 0) = \mathcal{OB}(n)$.

Table **??** brings together the common GPA steps and the geometric differences between orthogonal, oblique, and orthoblique rotation. It also distinguishes the quantities used to derive the projected gradient from the operations used to update $\mathbf{X}$.

Rotation constraints and standard errors

We already know how to impose constraints on $\mathbf{X}$ in orthogonal, oblique, and orthoblique rotation scenarios. A remaining matter is how these constraints are translated to the rotated loadings $\mathbf{\Lambda}$ and factor correlation matrix $\mathbf{\Psi}$ as this information is required for computing valid standard errors of rotated parameters.

Let $\boldsymbol{\theta}^*$ contain the r model parameters after factor rotation and $\mathbf{H}_\theta$ be the corresponding Hessian matrix. Define $\mathbf{C}_{\theta^*}$ as the $r \times c$ matrix whose columns contain the derivatives of each constraint with respect to the model parameters, where c denotes the number of rotation constraints. Then

$$\begin{pmatrix}\mathbf{H}_{\theta^*} & \mathbf{C}_{\theta^*}\\ \mathbf{C}_{\theta^*}^\top & \mathbf{0}\end{pmatrix}$$

is an augmented hessian matrix whose inverse can be used to obtain a variance-covariance estimate in maximum likelihood [Equation 17; Jennrich (1974)]. Alternatively, the augmented Hessian may be used in a sandwich estimator for other non-maximum likelihood methods

(Zhang, 2014).

We now show the hidden constraints over the rotated parameters and their corresponding derivatives. For orthogonal and oblique rotation, the derivatives of the constraints can be found elsewhere (Archer & Jennrich, 1973; Jennrich, 1973).

Let ζ be the function that transforms the rotation matrix into the rotated parameters. By the definition of gradient, at any tangent direction $\mathbf{V}$ we should have

$$\begin{aligned} \mathrm{D}f(\mathbf{X})[\mathbf{V}] &= \langle \Delta f_{\mathbf{\Lambda\Psi}}, \mathrm{D}\zeta(\mathbf{X})[\mathbf{V}] \rangle, \\ &= 0, \quad \text{at convergence}, \end{aligned} \tag{15}$$

where $\Delta f_{\mathbf{\Lambda\Psi}}$ is the gradient of the complexity function with respect to the rotated parameters. The second equality follows because at a given solution the complexity criterion must be stationary.

In orthoblique rotation, $\zeta(\mathbf{X}) = \{\mathbf{\Lambda} = \mathbf{\Lambda}_0\mathbf{X}^{-\top}, \mathbf{\Psi} = \mathbf{X}^\top\mathbf{X}\}$. The tangent space of the orthoblique manifold has the form $\mathbf{V} = \mathbf{X}^{-\top}\mathbf{\Omega}$, where the blocks of oblique factors in $\mathbf{\Omega}$ are arbitrary off-diagonal matrices, the block of orthogonal factors is skew-symmetric, and the intersections in $\mathbf{\Omega}$ between different blocks of factors i and j satisfy $\mathbf{\Omega}_{ij} = -\mathbf{\Omega}_{ji}^\top$ (i.e., they are arbitrary but skew-symmetric in the full $\mathbf{\Omega}$). Plugging this in Equation 15 and equaling to zero, we have

$$\begin{aligned} \mathrm{D}\zeta(\mathbf{X})[\mathbf{V}] &= \{-\mathbf{\Lambda}_0\mathbf{X}^{-\top}\mathbf{V}^\top\mathbf{X}^{-\top}, \mathbf{X}^\top\mathbf{V} + \mathbf{V}^\top\mathbf{X}\} \\ &= \{-\mathbf{\Lambda}(\mathbf{X}^{-\top}\mathbf{\Omega})^\top\mathbf{X}^{-\top}, \mathbf{X}^\top(\mathbf{X}^{-\top}\mathbf{\Omega}) + (\mathbf{X}^{-\top}\mathbf{\Omega})^\top\mathbf{X}\} \\ &= \{-\mathbf{\Lambda}\mathbf{\Omega}^\top\mathbf{\Psi}^{-1}, \mathbf{\Omega} + \mathbf{\Omega}^\top\}. \end{aligned}$$

Finally, substituting these expressions in Equation 15 and equaling to zero we have

$$\begin{aligned} 0 &= \langle \Delta f_{\mathbf{\Lambda}}, -\mathbf{\Lambda}\mathbf{\Omega}^\top\mathbf{\Psi}^{-1} \rangle + \langle \Delta f_{\mathbf{\Psi}}, \mathbf{\Omega} + \mathbf{\Omega}^\top \rangle, \\ &= \mathrm{trace}(-\Delta f_{\mathbf{\Lambda}}^\top\mathbf{\Lambda}\mathbf{\Omega}^\top\mathbf{\Psi}^{-1}) + \mathrm{trace}(\Delta f_{\mathbf{\Psi}}^\top(\mathbf{\Omega} + \mathbf{\Omega}^\top)), \\ &= \mathrm{trace}(-\mathbf{\Psi}^{-1}\Delta f_{\mathbf{\Lambda}}^\top\mathbf{\Lambda}\mathbf{\Omega}^\top) + \mathrm{trace}(2\Delta f_{\mathbf{\Psi}}\mathbf{\Omega}^\top). \end{aligned}$$

Therefore,

$$2\Delta f_{\mathbf{\Psi}} - \mathbf{\Psi}^{-1}\Delta f_{\mathbf{\Lambda}}^\top\mathbf{\Lambda} \tag{16}$$

must be symmetric, with the submatrices corresponding to the blocks of oblique factors being

also diagonal.

Following the orthoblique definition in (Equation 12), if p is the number of oblique factors in the first block, q is the number of oblique factors in the second block, and r is the number of completely orthogonal factors in the third block, then there is a total of $d = p + q + pq + pr + qr + r(r+1)/2$ independent constraints.

\textcolor{Calculate the derivatives of the d constraints in Equation 16. For the off-diagonal oblique constraints, I think the best way to present this is to write the differentials of Equation 16. For the oblique-oblique, oblique-orthogonal, and orthogonal-orthogonal constraints, calculate the differentials of Equation 16 minus its transpose.}

Estimation of the factor scores

To be done...

Software: orthoblique rotation with the latent package

A general rotation framework should allow researchers to compare and use different factor structures without developing a new optimization routine for each application. The `GPArotation` package already provides this framework for conventional orthogonal and oblique rotation, with a wide catalogue of rotation criteria. These are features retained in our `latent` package, with some additional capabilities. The relevant innovation here is the addition of the orthoblique manifold to restrict the factor-correlation structure as desired and the application of different rotation criteria to different blocks of factors.

Rotation in the `latent` package is performed through the `lrotate()` function, which can receive an initial loading matrix and factor correlation matrix to be rotated. An example is presented below:

```
library(latent)
# Orthogonal factor rotation with varimax:
fit_rotation <- lrotate(lambda = unrotated_lambda,
                        projection = "orth",
                        rotation = "varimax")
```

As it can be seen, the choice of criterion is separated from the choice of geometry: `rotation` specifies the criterion, whereas `projection` selects between orthogonal (`"orth"`), oblique (`"oblq"`), and orthoblique (`"poblq"`) manifolds. [7]

For the block structure developed in this paper, consider the following orthoblique target rotation:

```
library(latent)
# Orthoblique target rotation:
fit_rotation <- lrotate(lambda = unrotated_lambda,
                        projection = "poblq", oblique = c(3, 3),
                        rotation = "target", target = target)
```

Here, the `"poblq"` permits correlations within prespecified blocks of factors while retaining exact orthogonality between blocks, whereas `oblique` specifies the sizes of consecutive oblique blocks. For instance, with six factors, `oblique = c(3, 3)` gives two different oblique blocks of three factors with zero between-block correlations. If `unrotated_lambda` contained more than 6 columns, then any remaining factor would automatically be treated as orthogonal.

As an alternative to `lrotate()`, the function `lefa()` for exploratory factor analysis takes the raw data or a covariance matrix as inputs and performs factor extraction first and rotation later. `lefa()` allows all the same arguments as `lrotate()` but contains additional choices for factor extraction with either maximum likelihood (`"ml"`), unweighted least-squares (`"uls"`), or diagonally-weighted least-squares (`"dwls"`) for ordinal data.

```
# Oblique EFA with Maximum Likelihood and oblique oblimin rotation:
fit_efa <- lefa(data = data.matrix,
                sample.cov = NULL, # Optional covariance matrix
                nfactors = 6L,
                estimator = "ml",
                projection = "oblq",
                rotation = "oblimin")
```

Because different factor blocks may contain different levels of complexity, both `lrotate` and `lefa()` allow the application of different rotation criteria to arbitrary columns of $\mathbf{\Lambda}$. Suppose

[7] To call the orthoblique rotation we chose the label `"poblq"`, which stands for "partially oblique", because we did not found a convenient alternative that stands for "orthoblique". Options like `"orblq"` were considered but discarded for similarity with the oblique option.

that we have a block of three oblique factors and another block of three orthogonal factors. Then, we may choose to apply oblimin to the oblique block to maximize item simplicity but varimax to the orthogonal block to maximize the variance of the squared loadings within factors. This rotation specification can be easily implemented by providing a list of rotation criteria to the `rotation` argument:

```
# Orthoblique EFA with combined oblimin + varimax rotations:
fit_efa <- lefa(data = data.matrix,
                nfactors = 6L,
                estimator = "ml",
                projection = "poblq", oblique = 3,
                rotation = list(
                  oblimin = list(factors = 1:3),
                  varimax = list(factors = 4:6)))
```

Notice that we only had to specify a single number in the `oblique` argument to let GPA know that only the first three factors are indeed oblique and every other factor is orthogonal. Then, in the `rotation` argument the oblimin criterion is applied to the first three columns of $\mathbf{\Lambda}$, which corresponds to the oblique factors, whereas the varimax is applied to the last three columns, corresponding to the orthogonal factors.

The development version of `latent`, including the source and documentation of `lrotate()` and `lefa()` functions, is available at https://github.com/Marcosjnez/latent. You can install the package with the command `pak::pak("marcosjnez/latent")`.

Empirical applications of the orthoblique rotation

We now illustrate the use of the orthoblique manifold with two empirical applications in which neither a fully orthogonal nor a fully oblique rotation would represent the intended factor-correlation structure. The examples come from psychopathology and personality research. In both cases, we used target rotation, so the theoretical factor structure provides a starting point for the analyses, but we do not take these theoretical structures for granted. Instead, EFA will deliver a result that may agree or not with the target. In this regards, EFA becomes useful because the substantive structure provides information about the expected orientation

of the factors, but there is no reason to assume that all secondary loadings are exactly zero or that all the items are true indicators of their theoretical factors.

In the psychopathology example, we use a multitrait-multimethod (MTMM) design, where the same traits are measured using different informants. In these models, the associations between indicators may arise from the substantive traits but also from the method through which the measurements were obtained. Here, child, parent, and teacher reports constitute the different methods. EFA allows us to examine whether the expected trait structure emerges without fixing the loading pattern, whereas the orthoblique manifold keeps trait and method sources of common variance separated by imposing orthogonality between both blocks of factors.

In the personality example, the HEXACO hierarchy contains many items and factors. It is therefore plausible that some items have secondary loadings or that some facets are less clearly defined than expected. EFA allows these possibilities to be examined instead of assuming that every item belongs exclusively to its designated domain and facet. However, the distinction between general and specific factors still requires selected factor correlations to be zero. Thus, the exploratory model leaves all loadings free, whereas the orthoblique manifold constrains to zero the correlations between general and specific factors, as well as the correlations between specific factors belonging to different blocks.

Thus, both examples can be studied exploratorily while preserving the theoretically specified zero correlations between theoretically unrelated factors.

All analyses were conducted with the `latent` package and the `lefa()` function.

ASEBA. An exploratory Multitrait-multimethod model

The first application uses the MTMM data analyzed by Watts et al. (2022). The correlation matrix was obtained from the supplementary materials available at https://osf.io/hvskz/. We used the ABCD sample ($N = 2,119$), which includes child-, parent-, and teacher-reported psychopathology indicators measuring three traits: Attention Problems, Externalizing, and Internalizing. [8]

The original authors included the Attention Problems indicators in an Externalizing factor

[8]The original data included 56 indicators but two duplicated indicators (`ARGUc` and `ARGUp`) were identified after noticing the correlation matrix was not positive-definite. These indicators were removed from the supplied correlation matrix as we did not know to which method factor they would correspond, leaving 54 indicators available for the current analysis.

(Watts et al., 2022). This decision had a substantive motivation. Although the Achenbach System of Empirically Based Assessment (ASEBA) does not include Attention Problems in its Externalizing composite, Watts et al. (2022) included these indicators because attention-deficit/hyperactivity disorder is widely regarded as a marker of childhood Externalizing. They also referred to previous ASEBA analyses in which Attention Problems loaded strongly on an Externalizing factor. An exploratory analysis can examine whether the Attention Problems indicators are adequately captured by the Externalizing factor, as assumed in the merged specification.

Following the specification used by Watts et al. (2022), the attention-problems indicators were included with Externalizing. We therefore considered five exploratory factors: two trait factors corresponding to Externalizing and Internalizing, where the Externalizing label now includes the items of the Attention Problems factor, and three method factors corresponding to children, parents, and teachers.

Each indicator is expected to reflect both a substantive trait and a measurement method, but this allocation need not hold exactly in the data. Some indicators may show secondary loadings or may represent their designated trait or method less clearly than expected. An exploratory partially specified target rotation allows these possibilities to be examined without fixing non-target loadings to zero. At the same time, the MTMM structure requires trait factors to remain uncorrelated with method factors, while correlations may be allowed among trait factors and among method factors. Thus, the target criterion provides the substantive orientation of the factors, whereas the orthoblique manifold constrains all trait–method factor correlations to zero.

The two trait factors and the three method factors form two internally oblique blocks such that

$$\boldsymbol{\Psi} = \begin{pmatrix} \boldsymbol{\Psi}_T & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Psi}_M \end{pmatrix}.$$

Thus, the Externalizing-Internalizing correlation and the three correlations among method factors are estimated, whereas all six trait-method correlations are constrained to zero by the orthoblique manifold.

The model was fitted using unweighted least squares and target rotation as follows:

```
set.seed(2026)
R <- mtmm$ABCD$R # Extract the correlation matrix
```

```
efa <- lefa(sample.cov = R, sample.nobs = 2119,
            nfactors = 5, estimator = "uls",
            projection = "poblq", oblique = c(2, 3),
            rotation = "target", target = target)
```

The results showed that the estimated correlation between Externalizing and Internalizing was .251. The correlations were .298 between Children and Parents, .240 between Children and Teachers, and .302 between Parents and Teachers. In contrast, all six correlations between trait and method factors were zero to numerical precision as expected by the orthoblique constraints.

The partially specified target rotation produced a sparse loading pattern. Among the 162 loadings targeted toward zero, 160 (98.8%) had absolute values below .20, with a median absolute value of .030. The method factors were particularly well defined: all 54 expected method loadings exceeded .20 in absolute value, with a median absolute loading of .464. In contrast, the Externalizing indicators, apart from those assessing Attention Problems, loaded poorly on the merged trait factor. Of these 18 indicators, 16 had absolute loadings below .20, with a median of .079 and a range from .028 to .259. In contrast, 15 of the 18 Attention Problems indicators exceeded .20 on the same factor, with a median absolute loading of .433. Thereby, the original Externalizing indicators contributed little to the merged trait factor, which was mainly defined by Attention Problems.

Moreover, all 18 original Externalizing indicators loaded more strongly on their corresponding method factor than on the merged trait factor, with a median absolute method loading of .530. Therefore, their weak trait loadings do not mean that they contributed little to the entire solution. Rather, their shared variation was represented more strongly by the method factors, while the merged trait factor mainly reflected Attention Problems and not Externalizing.

Overall, this exploratory solution revealed that conflating the indicators of Externalizing and Attention problems in a single factor did not recover the intended broad representation of both sets of indicators. As such, retaining the Externalizing label for this merged factor was not appropriate as only the original indicators of Attention Problems loaded on the factor.

From a statistical perspective, the relevant feature of this example is that rotation criteria and restrictions on the factor correlation matrix are handled separately. Target rotation determines an interpretable orientation and permits empirically relevant cross-loadings,

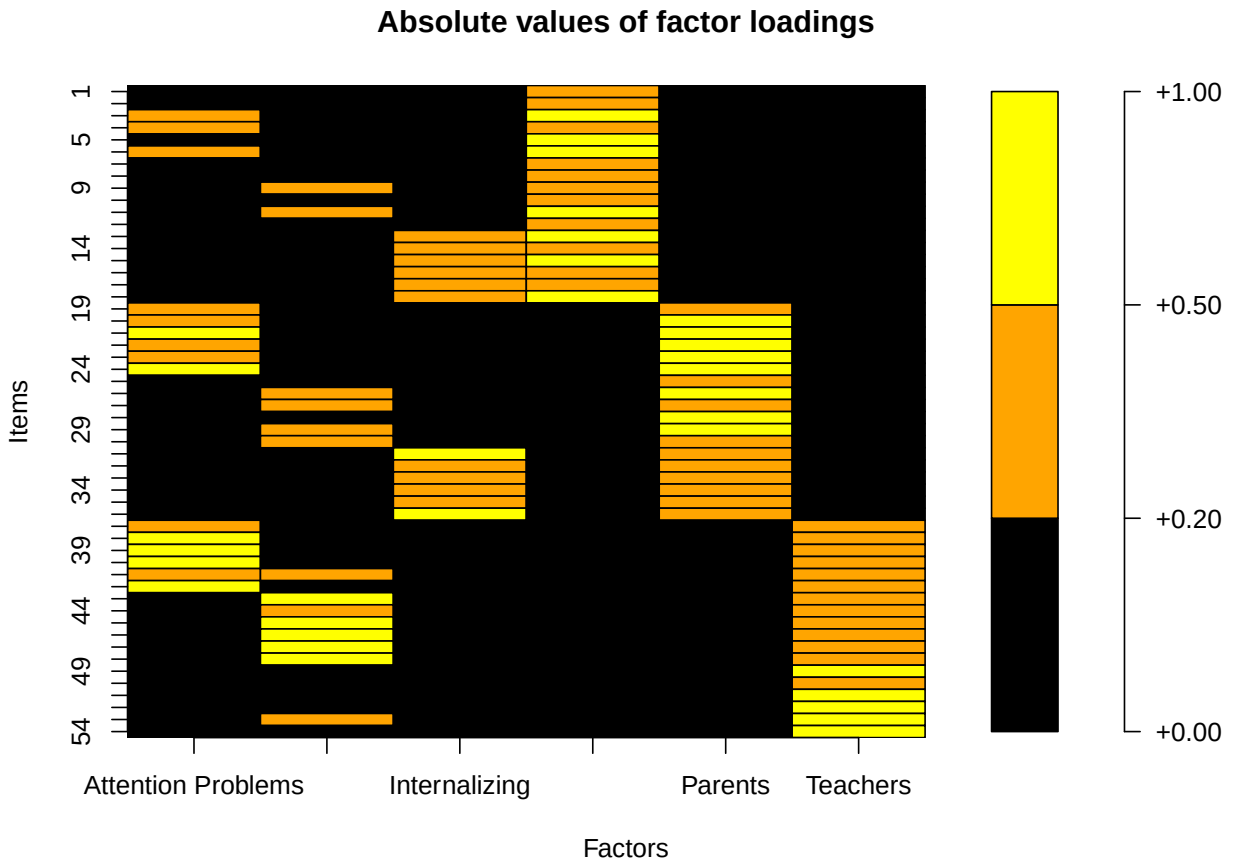


Figure 5: Absolute values of factor loadings for the multitrait-multimethod analysis.

whereas the orthoblique manifold guarantees the exact block-orthogonality structure that disentangles the variance explained by traits and methods. A conventional orthogonal rotation would have forced the correlations within the trait and method blocks to zero, while a conventional oblique rotation would have estimated the six trait-method correlations that must be excluded by the MTMM design. In both cases, the results would have been inconsistent with the theoretical model. Thus, the orthoblique manifold combined with target rotation made possible to test with EFA the adequacy of this MTMM analysis, revealing a possible modeling misspecification in Watts et al. (2022).

Finally, we also consider it important to mention that this use of target rotation is closely related to the extended target rotation proposed by Zhang et al. (2019) for MTMM data. Their method incorporates expectations about both the loading matrix and the factor correlation matrix into the rotation criterion (i.e., extended target rotation), so that selected trait-method correlations are rotated toward zero. Actually, the extended target rotation should be almost

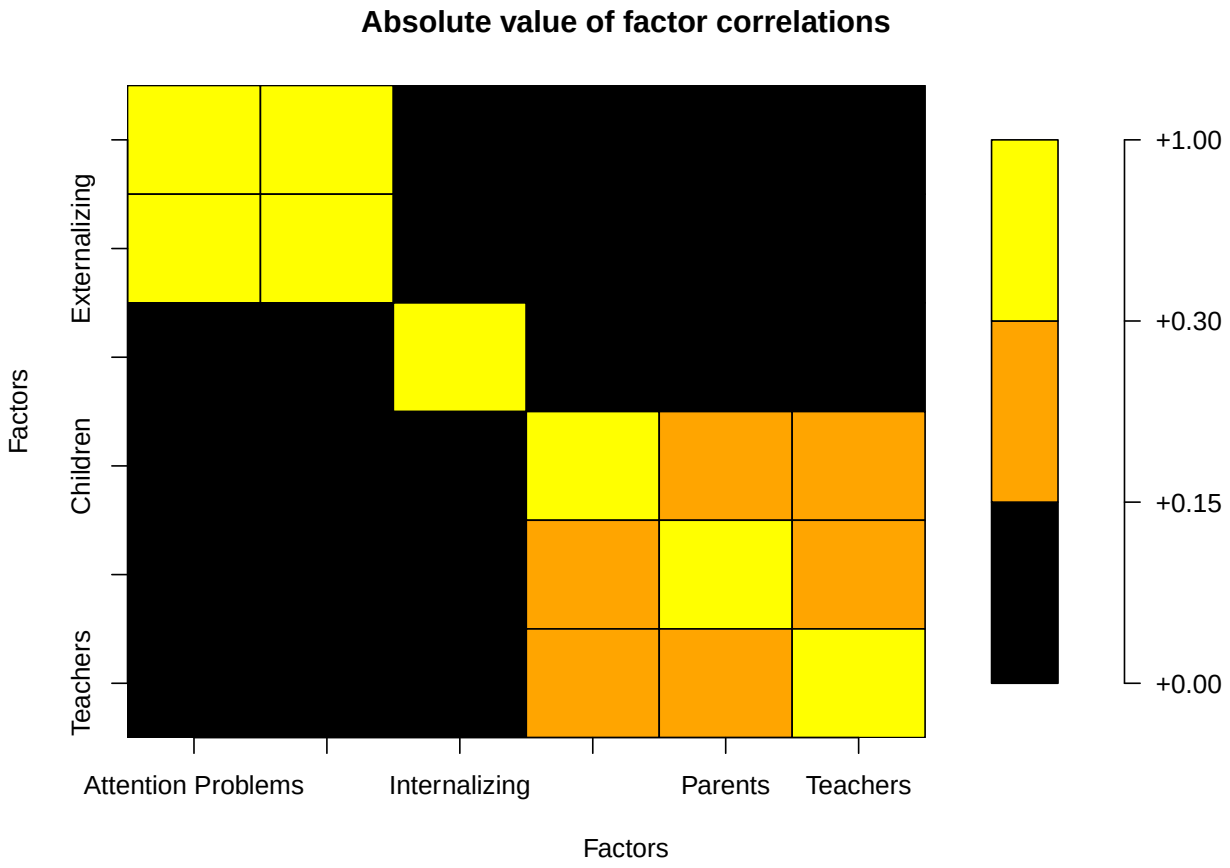


Figure 6: Absolute values of factor correlations for the multitrait-multimethod analysis.

equivalent to target rotation with the orthoblique manifold. However, the manifold and rotation roles are separated in our new approach: the target criterion is used only to orient the loading matrix, while the orthoblique manifold imposes the trait-method orthogonality exactly. Hence, the correlations encouraged to be small by extended target rotation in Zhang et al. (2019) are represented here as exact structural restrictions of the rotation manifold. As such, a benefit of our approach with regard to the extended target rotation is that the constant parameter that weights the relative contribution of target loadings and target correlations is not required.

HEXACO. A hierarchical mode with multiple oblique general factors

The second application uses the `icases` data obtained from the repository accompanying Anglim et al. (2022) at https://osf.io/72zp3/. We selected the MOOC sample ($N = 4,286$),

an international sample of participants who completed the HEXACO-100. The original study reports that the sample included participants from 131 countries, with 69% being female, and had a mean age of 34.7 years. For the present analysis, we used 96 HEXACO items corresponding to the six personality domains and their 24 facets, excluding the last four altruism items that do not belong to the HEXACO hierarchy (i.e, they are not expected to be nested in a general factor).

We fitted a 30-factor EFA model with six general factors, one for each HEXACO domain, and 24 specific factors. The first six factors were specified as one oblique block and the 24 specific factors were left orthogonal. The resulting factor correlation matrix has the form

$$\boldsymbol{\Psi} = \begin{pmatrix} \boldsymbol{\Psi}_G & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{24} \end{pmatrix},$$

where $\boldsymbol{\Psi}_G$ contains the correlations among the six domain-general factors. Therefore, the general factors can correlate, whereas every specific factor is orthogonal to the general factors and to all other specific factors. This manner, common variance would be partitioned between specific factors, representing narrow facet-level variance, and general factors, representing broader variance shared across facets.

This formulation extends the exploratory bifactor approach of Jennrich and Bentler (2011; 2012) to the case of multiple correlated general factors. In their biquartimin and bigeomin criteria, a single general-factor column is excluded from the simplicity rotation criterion, which is applied to the group-factor columns only. Here, there are six general-factor columns, and their orientation also has to be determined. We therefore applied a criterion to the general factor columns and another to the specific factor columns. Thus this example also illustrates the possibility of combining different rotation criteria within the same analysis.

To examine the extent to which the theoretical HEXACO structure is sound, there were many possible choices for analysis. Because there are 30 factors in total, a nontarget rotation has a large rotational search space and hence there exists the risk that we capitalize on chance. On the other hand, because the target matrix would constrain a large number of loadings toward zero, the analysis would become more confirmatory than intended. Therefore, four alternative specifications were fitted to explore how the recovered HEXACO structure varied across rotation criteria: geomin for the general and specific factors, oblimin for the general and specific factors, geomin for the general factors combined with target rotation for the

specific factors, and oblimin for the general factors combined with target rotation for the specific factors.[9] Because the specific-factor block was considerably larger and more complex than the general-factor block, target rotation was applied only to the specific factors in the target-based specifications, providing theoretical anchoring where rotational complexity was greatest. In the specifications without target rotation, the specific factors were rotated using the same rotation criterion as the general factors (i.e., oblimin or geomin). As such, these latter configurations can be seen as generalizations of the biquartimin and bigeomin criteria developed by Jennrich and Bentler (2011; 2012).

In the end, it was reassuring to see that the results of all the analyses converged overall. On the one hand, in both target-based specifications all 96 items had their largest absolute loading among the 24 specific factors on the theoretically-expected facet. When target rotation was not applied, this occurred for 72 of 96 items with geomin on the specific factors and for 73 of 96 items with oblimin on the specific factors. On the other hand, the correspondence with the expected facet structure was substantial even under the nontarget specifications: 75.0% of the items with geomin and 76.0% with oblimin had their largest absolute specific-factor loading on the corresponding facet. The broad recovery of the facet pattern was therefore not solely a consequence of applying a target. Nevertheless, this correspondence was not complete. Whereas the target specifications largely confirmed the expected HEXACO structure, the nontarget rotations may additionally provide insights into which facets and items may need further attention. Facets whose designated items have small loadings should be subjected to a closer examination of their content or item placement.

The general-factor loading patterns also broadly reflected the HEXACO domains across the four analyses. The item sets for Emotionality, Extraversion, Conscientiousness, Openness, and Agreeableness were recognizable in each solution, whereas Honesty–Humility was less consistently separated, especially in the target-based specifications. Some Honesty–Humility items showed salient cross-loadings on the Conscientiousness general factor, and additional general-factor cross-loadings remained. Thus, agreement across rotations was substantial but did not imply equally clear recovery of every domain. Because none of the four specifications targeted the general-factor loadings, this broad correspondence was not imposed directly by the target, although targeting the specific factors could still alter the partitioning of common variance.

[9] We set the oblimin parameter γ to 0 in all the analyses, so the criterion becomes quartimin. For geomin, we set $\epsilon = .01$ in all the analyses. For the analyses with the target rotation, the target used the known HEXACO facet memberships to anchor the orientation of the specific factors.

For brevity, we interpret further only the oblimin + target solution.[10] Here, oblimin was applied to the six general factors, leaving their loading pattern exploratory, whereas target rotation was applied to the 24 specific factors. Because the items are ordinal, the model was estimated with diagonally weighted least squares using polychoric correlations. The selected specification was fitted as follows:

```
rotation_model <- list(
  oblimin = list(gamma = 0, factors = 1:6),
  target = list(target = target[, 7:30], factors = 7:30)
)
efa <- lefa(data = mooc, nfactors = 30,
            ordered = TRUE, estimator = "dwls",
            projection = "poblq", oblique = 6,
            rotation = rotation_model)
```

The results showed that the absolute correlations among the six general factors ranged from .006 to .218, with a median of .107. All general-specific correlations and all off-diagonal correlations among the 24 specific factors were zero to numerical precision, as expected by the orthoblique constraints. Therefore, of the $30(29)/2 = 435$ possible factor correlations, only the $6(5)/2 = 15$ correlations among the general factors were estimated, while the remaining 420 were exactly zero.

The six general factors remained substantially exploratory in this analysis. The median absolute loading on the theoretically-expected general factor was .385; 85 of the 96 designated general-factor loadings exceeded .20 in absolute value, and 430 of the 480 general-factor cross-loadings (89.6%) were below .20. For the 24 specific factors, the median absolute theoretically expected loading was .430, and 91 of these 96 loadings exceeded .20. In contrast, 2,205 of the 2,208 specific-factor cross-loadings (99.86%) were below .20, and no cross-loading exceeded .30.

The conceptual advantage of the orthoblique manifold was particularly clear in this example. A fully oblique 30-factor rotation would allow all 435 factor correlations, although only the 15 correlations among the six general factors are needed for this representation. Conversely, a fully orthogonal rotation would force these 15 correlations to zero. At the same time, the mixed rotation criterion leaves the six general factors under a less restrictive simplicity

[10] To check the other solutions, see the repository https://github.com/Marcosjnez/orthoblique.

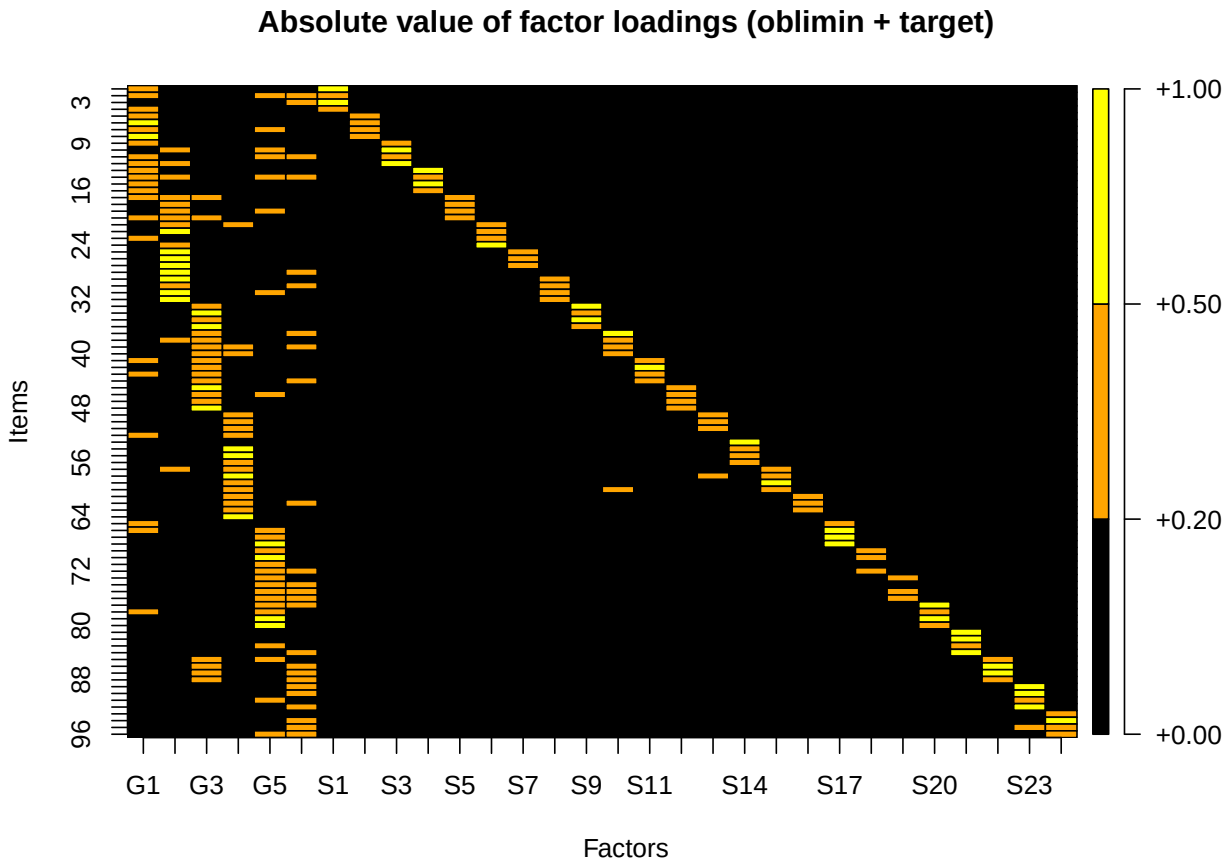


Figure 7: Absolute values of factor loadings for the HEXACO oblimin + target solution.

criterion and uses the target only where the large number of specific factors makes theoretical anchoring desirable, reducing the risk of capitalizing on chance.

Taken together, these applications illustrate the distinction between rotation criteria and manifold restrictions. Complexity criteria can yield different sparsity and parsimony patterns for each block of factors whereas the orthoblique manifold defines the admissible factor-correlation geometry exactly. The two mechanisms can therefore be combined without turning the factor model into a confirmatory simple-structure model.

Discussion

Until now, conventional GPA-based factor rotation has generally treated factors as either completely orthogonal or completely oblique. We developed the orthoblique rotation to allow

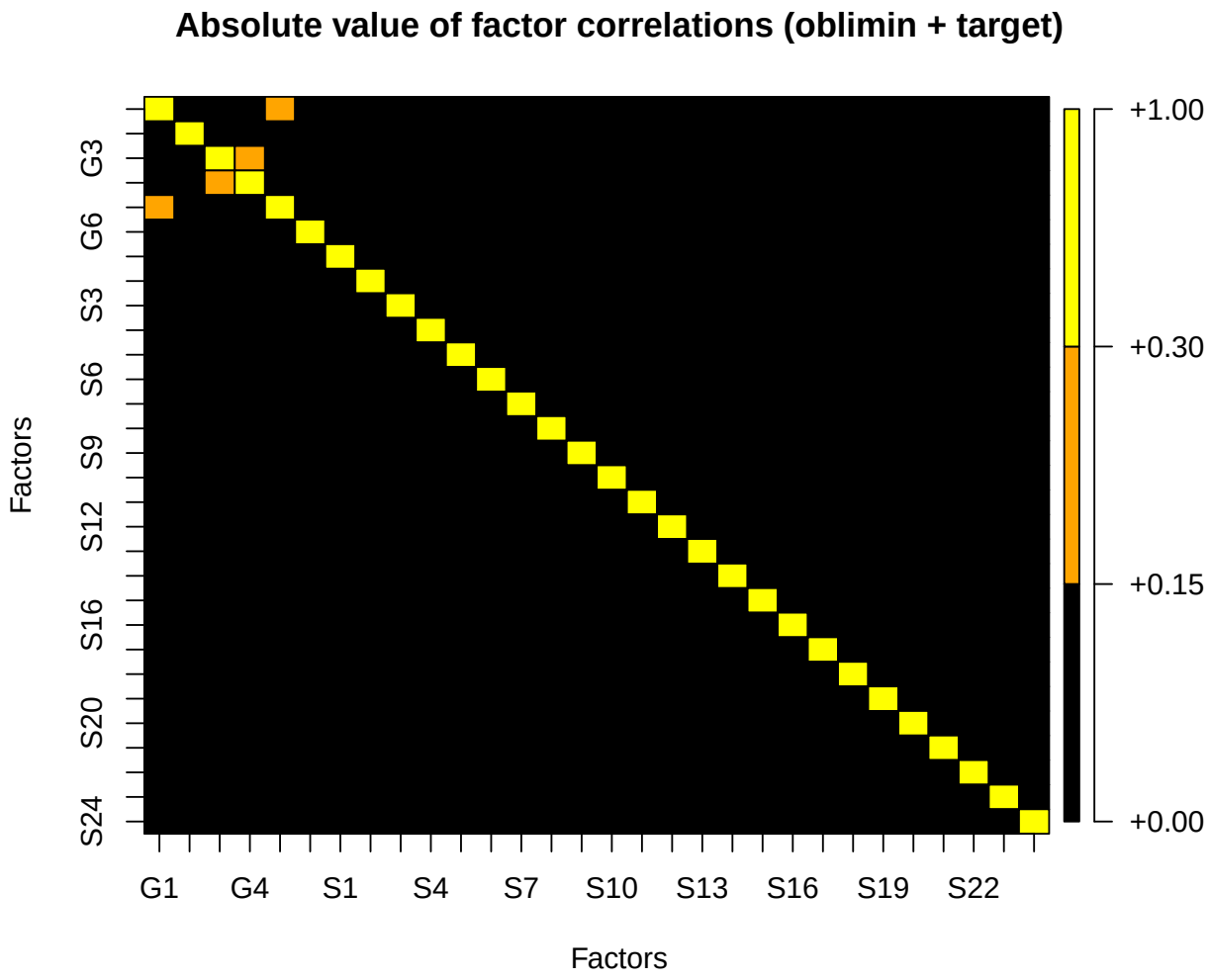


Figure 8: Absolute values of factor correlations for the HEXACO oblimin + target solution.

oblique blocks of factors to coexist with exact orthogonality between selected blocks. The empirical applications illustrated two complementary uses of this geometry: an MTMM representation in which trait and method factors can correlate within their respective blocks while remaining orthogonal between blocks, and a general-specific personality representation in which several general factors may correlate while remaining orthogonal to the specific factors, which are also mutually orthogonal.

We also provided a detailed and self-contained review of the GPA, connecting its computational steps with the geometry of the rotation constraints. In particular, starting from the defining constraints, we derived the tangent and normal spaces and used their orthogonal decomposition to establish the Riemannian gradient for orthoblique rotation. The

conventional orthogonal and oblique formulas can be recovered within the same account, so orthoblique rotation can be regarded as a generalization of orthogonal and oblique rotations. Therefore, the orthogonal and oblique manifolds are not competing constructions that must be replaced, but special cases of the more general orthoblique manifold. We hope that these explicit connections make factor rotation a less obscure matter in factor analysis and help researchers understand how further extensions can be constructed.

In the end, this extension should be understood as a continuation of the general rotation program developed by Jennrich (2001), Jennrich (2002), and Bernaards and Jennrich 2005. Moreover, the empirical analyses on data with multiple general factors illustrated that the current work can also be viewed as a continuation of the bifactor exploratory analysis methods with biquartimin and bigeomin (Jennrich & Bentler, 2011, 2012). A central insight of Jennrich was that the same algorithm could optimize many simplicity criteria. Here, we showed how the same geometric reasoning can also accommodate different patterns of factor-correlation constraints.

Additionally, we provided a new R package to exploit these new rotation capabilities. `lefa()` combines factor extraction and `lrotate()` in a single call. This makes the same orthoblique specification available when the analysis starts from raw data or a sample covariance matrix. Through the `projection`, `oblique`, and `rotation` arguments, researchers can specify the factor-correlation blocks and the criteria applied to their loadings within the same analysis. Consequently, the proposed extension is already available for use.

We hope that the combination of the orthoblique manifold with mixed rotation criteria considerably expands the range of possible rotation specifications. Different assumptions about factor correlations can be combined with different simplicity criteria for the corresponding loading blocks. The present applications illustrate only a few of these combinations; we have not examined the many other possibilities or established which will be most useful in different substantive settings. The `latent` package makes these unexplored combinations already available to researchers, who can investigate new rotation configurations based on their parameter recovery, stability, and interpretability. In this sense, the current work provides both a geometric construction and a software basis for further research on factor rotation and, more broadly, on EFA.

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